



IWATSU

Semiconductor Curve Tracer

Digital Oscilloscope

Analog Oscilloscope

Isolation Measurement System

XY Monitor and Multi-Window Waveform Monitor

Digital Multimeter

Universal Counter

Function Generator

Delay Pattern Generator

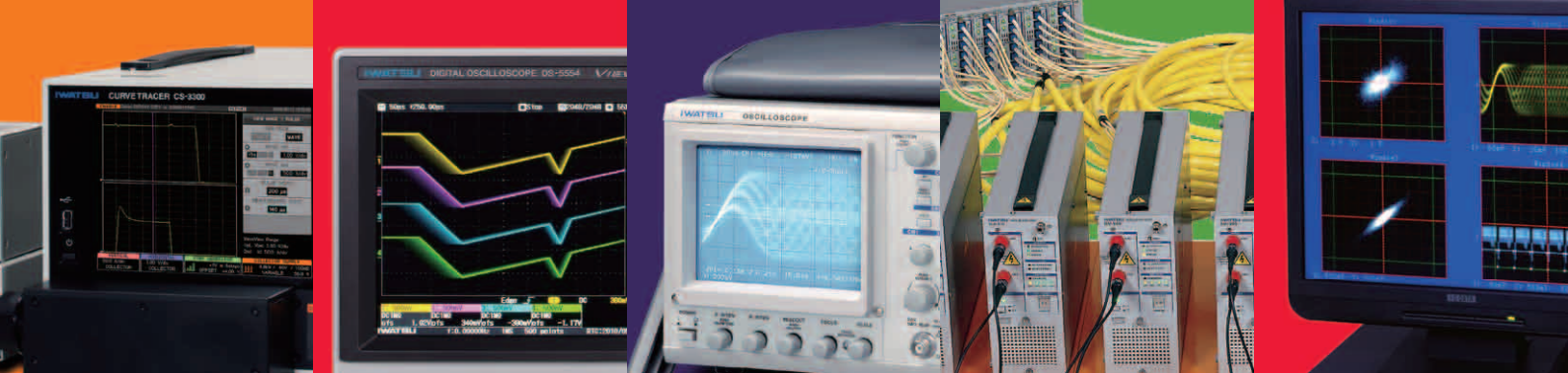
B-H Analyzer

Targeting tomorrow's electronics

At IWATSU, our focus is always on the future. With the relentless pace of development in the electronics industry, success demands innovation, creativity, and an unwavering commitment to research and development. Building on our solid base of accumulated basic research, we are expanding our cutting-edge R&D with high technology both domestically and overseas. In addition to power electronics and its managements such as inverters for train, PV(photovoltaics), etc. we manufacture a wide range of electronic equipment and systems to cover various types of demands from industries and research for energy-efficient power managements.



1930's	1938	<i>Iwatsu Electric Co., Ltd. founded in Shibuya, Tokyo.</i>
1950's	1952	<i>Grant-in-aid for industrial technology research was offered to our design of shock-wave measurement device. Two years later, Japan's first domestic oscilloscope was put on the market (trademark registered as SYNCHROSCOPE).</i>
	1957	<i>Listed on the first section of the Tokyo Stock Exchange.</i>
1960's	1961	<i>Development and manufacture of proprietary CRT for waveform observation started.</i>
	1962	<i>Development and manufacture of proprietary IC started.</i>
1970's	1970	<i>The first domestic IC oscilloscope released, providing a compact and light oscilloscope.</i>
	1974	<i>Colona-Denshi Co., Ltd., (present name: Iwatsu Test Instruments Corporation, Aizu factory) was established in Wakamatsu, Aizu, Fukushima as a production base for electric measurement equipment.</i>
1980's	1980	<i>World's fastest analog storage oscilloscope released.</i>
1990's	1991	<i>An overseas affiliate Iwatsu (Malaysia) Sdn. Bhd. (presently a consolidated subsidiary of Iwatsu Electric Co., Ltd.) was established.</i>
	1999	<i>Digital oscilloscopes were joint-developed with LeCroy Corporation.</i>
2000's	2000	<i>Iwatsu TME Service Co., Ltd., (present name: Iwatsu Test Instruments Corporation), a service company specializing in measurement equipment, was established.</i>
	2002	<i>Iwatsu Test Instruments Corporation was established from the measurement division of Iwatsu Electric Co., Ltd. The world's only 1GHz bandwidth analog storage oscilloscope, TS-81000 was released, featuring high speed high brightness writing.</i>
	2004	<i>50th anniversary of oscilloscope sales. Digital oscilloscopes to support Microsoft® Windows® OS were released. Digital multi-meter with two-channel input, VOAC7520 was released.</i>
	2005	<i>Full-scale entry into the field of measurement for the automobile industry.</i>
	2006	<i>Four models of digital oscilloscope DS-5100 series were released, providing high performance with low cost. Full-scale entry into the field of measurement for high performance electromagnetic steel sheets with the world's first V-H analyzer IE-1131B.</i>
	2007	<i>An isolation system for power electronics, DM-8000 was released, providing highly accurate measurement of ultra-high voltages.</i>
	2009	<i>Full-scale entry into the field of measurement for power semiconductors, with three models of CS-3000 series, a semiconductor curve tracer supporting high current at 1,000A. Capacitance displacement meter with high resolution and high stability, the ST-3541 series were released.</i>
	2011	<i>Two models of CS-10000 series, a semiconductor curve tracer supporting ultra-high voltage high current, and three models of CS-5000 series were released, providing support to all needs in the field of measurement for power semiconductors. Genuinely domestic highly accurate measurement equipment, radiation dosimeter SV-1000/SV-2000 were released.</i>



Semiconductor Curve Tracer

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CS-3300, CS-3200, CS-3100, CS-5300, CS-5200, CS-5100, CS-10800, CS-10400

Digital Oscilloscope

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DS-5554, DS-5552, DS-5534, DS-5532, DS-5524, DS-5522, DS-5514, DS-5512

Analog Oscilloscope

12

SS-7847A, SS-7840A, SS-7830A, SS-7840AP, SS-7830AP

Isolation Measurement System

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DM-8000

XY Monitor and Multi-Window Waveform Monitor

19

SV-1014

Digital Multimeter

20

VOAC 7523H, VOAC 7520H, VOAC 7522H, VOAC 7521H

Universal Counter

24

SC-7207H, SC-7206H, SC-7205H, SC-7217

Function Generator / Boost Amp

28

SG-4105, SG-4104 / SG-300

Delay Pattern Generator

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DG-8000

B-H Analyzer

32

SY-8218, SY-8219



Semiconductor Curve Tracer

CS-3000 Series

- Maximum peak voltage: 3,000V (high-voltage mode for all models)
- Maximum peak current: 1,000A (CS-3300 high-current mode)
- All models support the leakage mode (cursor resolution: 1pA)
- USB for Screen copying and setup saving
- LAN interface for remote control

*All models of CS-3000 series are equipped with Fixture, CS-301(Fixture S) for CS-3100 and CS-302(Fixture M) for CS-3200/CS-3300 as standard.



CS-3100



CS-3200 / CS-3300



Semiconductor Curve Tracer

CS-5000 Series



CS-5300

CS-5300 5,000V / 1,000A with HC mode
CS-5200 5,000V / 400A with HC mode
CS-5100 5,000V / 15A

Best suited for measuring the breakdown of a power device which has 3,300V withstand voltage

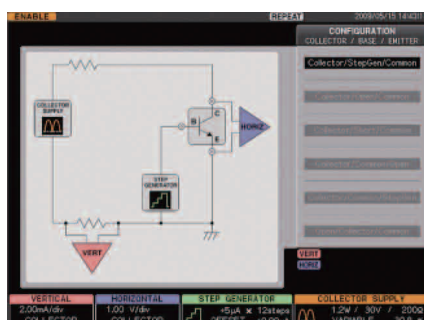
- Maximum peak voltage: 5,000V (high-voltage mode for all models)
- Maximum peak current: 1,000A (CS-5300 high-current mode)
- The CS-5000 Series achieved 5,000V of max. peak voltage with the same size as the CS-3000 Series.

*All models of CS-5000 Series are equipped with the CS-303 (Fixture M) as standard.

Lineup

Items	Product name	Model number	Remarks
Main unit	Semiconductor curve tracer	CS-3300	3,000V / 1,000A with HC mode - CS-302 / CS-500
	Semiconductor curve tracer	CS-3200	3,000V / 400A with HC mode - CS-302 / CS-500
	Semiconductor curve tracer	CS-3100	3,000V / without HC mode - CS-301 / CS-500
	Semiconductor curve tracer	CS-5300	5,000V / 1,000A with HC mode - CS-303 / CS-500
	Semiconductor curve tracer	CS-5200	5,000V / 400A with HC mode - CS-303 / CS-500
	Semiconductor curve tracer	CS-5100	5,000V / without HC mode - CS-303 / CS-500
Fixture	Fixture S	CS-301	Standard for CS-3100.
	Fixture M	CS-302/CS-303	CS-302 standard for CS-3200, CS-3300 and CS-303 standard for all CS-5000 series
Test adapter	Test adapter	CS-500	A test adapter is supplied with the main unit for both CS-301, CS-302 and CS-303.
	T0-type test adapter	CS-501	Socket suited to the T0-type package (with three terminals) for both CS-301, CS-302 and CS-303.
Alligator clip	Small alligator clip (red) x 10	CS-001	Option for CS-302/CS-303 fixture M
	Small alligator clip (black) x 10	CS-002	Option for CS-302/CS-303 fixture M
Software options	Semiconductor parameter search	CS-800	A software option to be installed in the CS-3000 series and CS-5000 series.
	Semiconductor parameter measurement	CS-810	To be used on the (optional) PC when the CS-800 is installed in the main unit.

User-friendly measurement screen



Graphical configuration selection



Transistor I-V characteristic example (TRACE mode)



Vbe and Ic waveforms in the High-current pulse mode (WAVE mode)

Support for a maximum peak voltage of 5,000V and a maximum peak current of 1,000A



1 Display

The 8.4-inch clear screen displays the waveform details. This large screen displays the menu and parameters without limiting the waveform grid.

2 Readout

The readout display, which is color-coded by function, displays operation settings in an easy-to-understand way.

3 Connectivity

The CS-5000 and CS-3000 series features a USB port on the front panel facilitating report creation. Screen images can be saved to a USB device by pressing the COPY button.

4 Rapid activation

5 Breaker

Breakers for Enabling/Disabling high voltage and high current (CS-5300/5200, CS-3300/3200) are provided on the front panel.

6 Menu operation section

Use the function knob and function keys to operate the menu.

7 STEP GENERATOR

The step setting and offset can be manipulated using this dedicated knob.

8 COPY button

9 CONFIGURATION button

This shows the connection and signal application status.

10 VIEW/PULSE button

This is used to switch between the TRACE and WAVE views.

In the WAVE view, the screen displays the time axis waveform of the applied voltage and current, which conventional curve tracers could not observe.

11 CURSOR button

DOT CURSOR	Vert, Horiz, β or gm
FLINE CURSOR	Vert, Horiz, 1/grad, intercept
FREE CURSOR	Vert, Horiz, β or gm
WINDOW CURSOR	Vert, Horiz of WINDOW area, β or gm.

12 VERTICAL and HORIZONTAL knobs

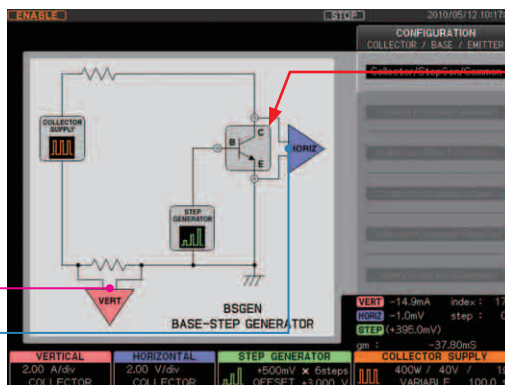
These dedicated knobs are located in an easily accessible position.

13 VARIABLE knob

This is used to adjust the output voltage of the collector supply. A durable knob is employed due to frequent use.

Observation of I-V curves as well as applied waveforms of voltage and current

Application of the current and voltage pulses in the high current mode



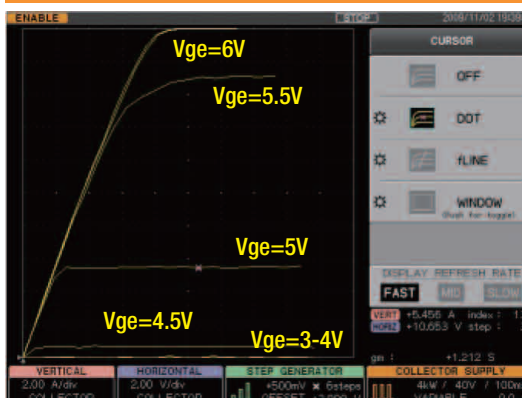
The pulse widths of the applied current and voltage and the measurement point can be specified in a range between 50μs and 400 μs (CS-3200 and CS-3300).

Example of measuring breakdown voltage



The SWEEP TYPE can be selected from DOWN, UP, and CUSTOM (all models). With CUSTOM specified, the range between specified values can be swept.

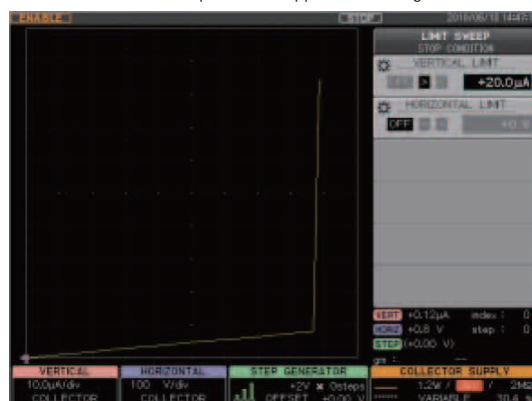
Example of measuring MOSFET "I-V (current - voltage)" characteristics



Typical I-V characteristics measurement on MOSFET

CS-800 semiconductor parameter search (optional)

The main unit software option that supports the voltage and current limit function, as well as the automatic Vth measurements.



LIMIT SWEEP

If the voltage and current limit values are specified before SWEEP operations, the operations stop at the next measurement point after the limit value has been exceeded.



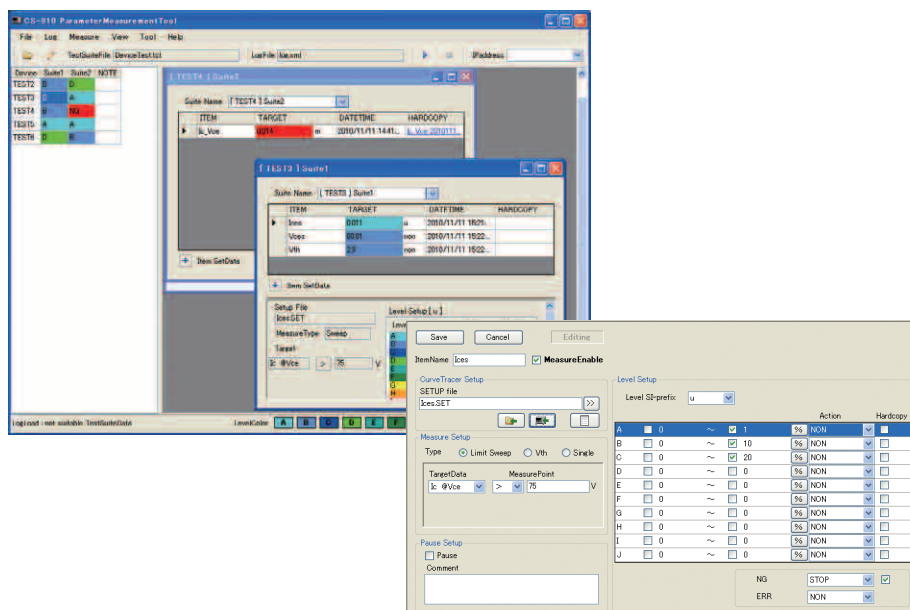
Vth and hFE SETUP

Vth and hFE can be automatically sought and measured.

Example of automatic Vth measurement

CS-810 semiconductor parameter measurement (optional)

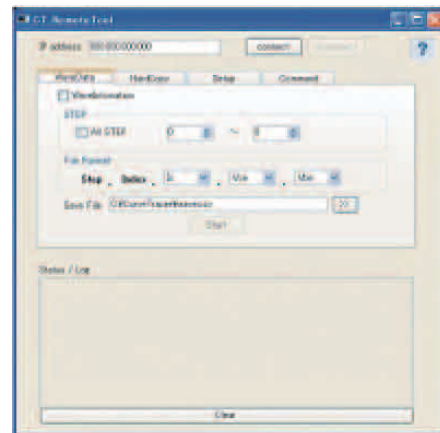
This is a PC application used to connect the main unit incorporating CS-800 and an (optional) PC by a LAN. This application automatically performs measurements by applying the measurement conditions specified using the PC to determine whether the data is appropriate.



Remote tool

If the use of USB memory is not permitted for security reasons, a copy of the screen, measurement results in the CSV format, and other data can be directly transferred to the PC via LAN.

*The Remote tool can be downloaded from the Iwatsu website (freeware)



Accessories

Standard accessories for CS-3100



When the CS-501 is mounted.

*CS-501 is optional.

Standard accessories for CS-3300 and CS-3200



When the patch panel removed

CS-500 test adapter

CS-302 fixture M*

Wire set

*CS-303 for CS-5300, CS-5200, CS-5100

CS-500 test adapter

The CS-500 Test adapter is supplied with the product. Use this adapter to connect the prepared jig to the fixture.



CS-501 TO type test adapter

This socket is suited to the TO type package (with three terminals). The supported lead distance is from 1.52 to 4.57mm.



CS-001 (red x 10pcs.), CS-002 (black x 10pcs.) Small alligator clips

These small alligator clips can be attached to the wire set supplied with the CS-302 fixture.



CS-001 CS-002

Power Curve Tracer CS-10000 Series

CS-10800/CS-10400 Custom Order

Best suited for measuring the characteristics of power devices

•Ultra-high voltage and high current

The following HV and HC functions are added to the CS-3100

High voltage mode	10kV(+DC only) 8,000A(CS-10800)
High current mode	4,000A(CS-10400) Pulse width/Pulse interval/ Measuring point variable



10 kV curve tracer Custom Order

Ideal for measuring the withstand voltage characteristic of high voltage diode thyristors

Output section	
Voltage waveform	Commercial power supply half-wave rectification waveform
Maximum voltage	10kV at peak (without load)
Maximum current	100mA at peak
Display section	
Voltage axis sensitivity	50V/div to 1,000V/div, 5 ranges at 1-2-5 step
Current axis sensitivity	0.1mA/div to 10mA/div, 7 ranges at 1-2-5 step



Semiconductor Curve Tracer Series Specifications

Model		CS-3300		CS-3200		CS-3100	
Collector supply							
Mode	High voltage1	AC, ± Rectified SIN, ± DC, ± LEAKAGE					
	High voltage2	—					
	High current	Pulse only				—	
Maximum peak power	High voltage1	120mW, 1.2W, 12W, 120W, 390W *390 W can be selected excluding the maximum peak voltage setting at 3,000V range.					
	High voltage2	—					
	High current	400W, 4kW, 10kW		400W, 4kW		—	
High voltage mode1			Maximum peak current (maximum peak pulse current)		Maximum peak voltage		
			75mA(150mA)		3,000V(2,500V / AC)		
			750mA(1.5A)		300V		
			7.5A(15A)		30V		
Loop compensation		The stray capacitance between the collector terminal and the ground of the fixture is compensated by the hardware in the high voltage mode. Digital compensation is also available.					
High voltage mode2		—					
High current mode (pulse only)							—
	Maximum peak current	Maximum peak voltage	Maximum peak current	Maximum peak voltage			
	1,000A	40V	400A	40V			
	400A	40V	40A	40V			
	40A	40V					
Pulse width / measurement point		The pulse width is variable in the range from 50μs to 400 μs. Measurement points can be specified (with a resolution of 10 μs/step).				—	
Maximum data points		20 to 1,000 points can be specified per trace.					
MEASUREMENT MODE		REPEAT, STOP/SINGLE, SWEEP					

Step generator

Current mode	Amplitude range: 50nA to 200mA, 21 ranges, 1-2-5 step Maximum current: $\pm$ 2A / Offset: $\pm$ 10 times STEP AMPLITUDE setting	
Voltage mode	Amplitude range: 50mV to 2V, 6 ranges, at 1-2-5 step Maximum voltage: $\pm$ 40V / Offset: $\pm$ 10 times STEP AMPLITUDE setting	
Step rate	Staircase wave: 2times of 50Hz or 60Hz (50Hz or 60Hz in the AC mode) Pulse : The pulse changes in the range from 80ms to 1,000 ms. (The lowest frequency limited by the maximum peak power consumption setting.)	Staircase wave: 2times of 50Hz or 60Hz (50Hz or 60Hz in the AC mode)
Pulse step	The pulse width is variable in the range from 50 μ s to 400 μ s at 10 μ s resolution.	
Number of steps	0 to 20steps	

AUX output

Range	OFF, -40 to +40V, variable with at 100mV resolution
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Vertical axis

Collector current	High voltage	1 μ A/div to 2A/div, 20 ranges, at 1-2-5 step		
	High current	100mA/div to 100A/div, 10 ranges, at 1-2-5 step	100mA/div to 50A/div, 9 ranges, at 1-2-5 step	—
Emitter current(LEAKAGE)	1 nA/div - 2 mA/div, 20 ranges, at 1-2-5 step			

Horizontal axis

Collector voltage	High voltage	50mV/div to 500V/div, 13 ranges, at 1-2-5 step 50mV/div to 1,000V/div, 14 ranges, at 1-2-5 step	
	High current	50mV/div to 5V/div, 7 ranges, at 1-2-5 step	—
Base / emitter voltage		50mV/div to 5V/div, 7 ranges, at 1-2-5 step	

Miscellaneous

Display	8.4-inch color TFT-LCD (SVGA 800 x 600 pixels)				
Data save / recall	Internal: Memory (setup: 256, REF waveform: 4) External: Removable storage connected to the USB port (setup, waveform save/recall, screen hard copy of display (BMP, TIFF, PING))				
USB	1 port (USB1.1) Setup, Waveform Save/Recall, Hard copy of display (BMP, TIFF, PING)				
Remote control	Remote control by LAN: 1 port (100BASE-TX)				
Power supply	AC 100V - AC240V, single phase, 50Hz/60Hz, Approx. 500VA				

Mechanism section

Dimensions (mm)	424 (W) x 555.2 (L) x 354.5 (H) (without the external projection portion)		424 (W) x 555.2 (L) x 221 (H) (without the external projection portion)	
Weight	Approx. 45kg (without optional accessories)		Approx. 30kg (without optional accessories)	

Model	CS-5300		CS-5200		CS-5100	CS-10800		CS-10400	
Collector supply									
Mode	High voltage1	AC, ± Rectified SIN, ± DC, ± LEAKAGE							
	High voltage2	—				+DC			
	High current	Pulse only			—		Pulse only		
Maximum peak power	High voltage1	120mW, 1.2W, 12W, 120W, 390W *120mW and 390 W can be selected excluding the maximum peak voltage setting at 5,000V range.				120mW, 1.2W, 12W,120W, 390W *390 W can be selected excluding the maximum peak voltage setting at 3,000V range.			
	High voltage2	—				4kW			
	High current	400W, 4kW, 10kW		400W, 4kW		—		600W, 6kW, 60kW, 64kW	
High voltage mode1	Maximum peak current (maximum peak pulse current)		Maximum peak voltage		—	Maximum peak current (maximum peak pulse current)		Maximum peak voltage	
	25mA(25mA)		5,000V(2,500V / AC)			75mA(150mA)		3,000V(2,500V / AC)	
750mA(1.5A)		300V		750mA(1.5A)		300V			
7.5A(15A)		30V		7.5A(15A)		30V			
Loop compensation	The stray capacitance between the collector terminal and the ground of the fixture is compensated by the hardware in the high voltage mode. Digital compensation is also available.								
High voltage mode2	—					Maximum peak current		Maximum peak voltage	
						400mA		10,000V	
High current mode (pulse only)	Maximum peak current Maximum peak voltage 1,000A 40V 400A 40V 40A 40V		Maximum peak current Maximum peak voltage 400A 40V 40A 40V		—	Maximum peak current Maximum peak voltage 8,000A 60V 4,000A 60V 400A 60V 40A 60V		Maximum peak current Maximum peak voltage 4,000A 60V 400A 60V 40A 60V	
	Pulse width / measurement point		The pulse width is variable in the range from 50μs to 400μs. Measurement points can be specified (with a resolution of 10μs/step).				—		The pulse width is variable in the range from 50μs to 400μs. (50μs to 1,200μs for 8kA range) Measurement points can be specified at 10μs/step. (50μs to 900μs for 4kA range)
Maximum data points	20 to 1,000 points can be specified per trace.								
MEASUREMENT MODE	REPEAT, STOP/SINGLE, SWEEP								

Step generator

Current mode	Amplitude range: 50nA to 200mA, 21 ranges, 1-2-5 step Maximum current: $\pm$ 2A / Offset: $\pm$ 10 times STEP AMPLITUDE setting		
Voltage mode	Amplitude range: 50mV to 2V, 6 ranges, at 1-2-5 step Maximum voltage: $\pm$ 40V / Offset: $\pm$ 10 times STEP AMPLITUDE setting		
Step rate	Staircase wave: 2times of 50Hz or 60Hz (50Hz or 60Hz in the AC mode) Pulse : The pulse changes in the range from 80ms to 1,000ms. (The lowest frequency limited by the maximum peak power consumption setting.)	Staircase wave: 2times of 50Hz or 60Hz (50Hz or 60 Hz in the AC mode)	Staircase wave: 2times of 50Hz or 60Hz (50 μ s or 60Hz in the AC mode) Pulse : The pulse changes in the range from 80ms to 1,000ms. (The lowest frequency limited by the maximum peak power consumption setting.)
Pulse step	The pulse width is variable in the range from 50 μ s to 400 μ s at 10 μ s resolution.		
Number of steps	0 to 20steps		

AUX output

Range	OFF, -40 to +40V, variable with at 100mV resolution
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Vertical axis

Collector current	High voltage	1 μ A/div to 2 A/div, 20 ranges, at 1-2-5 step		
	High current	100mA/div to 100 A/div, 10 ranges, at 1-2-5 step	100mA/div to 50 A/div, 9 ranges, at 1-2-5 step	—
Emitter current(LEAKAGE)		1nA/div - 2mA/div, 20 ranges, at 1-2-5 step		

Horizontal axis

Collector voltage	High voltage	50mV/div to 500V/div, 13 ranges, at 1-2-5 step	50mV/div to 1,000 V/div, 14 ranges, at 1-2-5 step	50mV/div to 1,000V/div, 14 ranges, at 1-2-5 step
	High current	50mV/div to 5V/div, 7 ranges, at 1-2-5 step	—	50mV/div to 10V/div, 8 ranges, at 1-2-5 step
Base / emitter voltage		50mV/div to 5V/div, 7 ranges, at 1-2-5 step		

Miscellaneous

Display	8.4-inch color TFT-LCD (SVGA 800 x 600 pixels)		
Data save / recall	Internal: Memory (setup: 256, REF waveform: 4) External: Removable storage connected to the USB port (setup, waveform save / recall, screen hard copy)		
USB	1 port (USB1.1) Setup, Waveform Save/Recall, Hard copy of display (BMP, TIFF, PING)		
Remote control	Remote control by LAN: 1 port (100BASE-TX)		
Power supply	AC100V - AC240V, single phase, 50Hz/60Hz, Approx. 500VA		AC100V - AC240V, single phase, 50Hz/60Hz

Mechanism section

Dimensions (mm)	424 (W) x 555.2 (L) x 354.5 (H) (without the external projection portion)	424 (W) x 555.2 (L) x 221 (H) (without the external projection portion)	1,110 (W) x 1,150 (L) x 1,216 (H) Dimensions (mm) (without the external projection portion)
Weight	Approx. 45kg (without optional accessories)	Approx. 30kg (without optional accessories)	Approx. 390kg

Digital Oscilloscope New DS-5500 Series VIEWGO II

3-year warranty
when registered as a web user

GPIO
[Option]



500MHz 4ch 2GS/s 1M point	DS-5554
500MHz 2ch 2GS/s 1M point	DS-5552
350MHz 4ch 2GS/s 1M point	DS-5534
350MHz 2ch 2GS/s 1M point	DS-5532
200MHz 4ch 2GS/s 1M point	DS-5524
200MHz 2ch 2GS/s 1M point	DS-5522
100MHz 4ch 1GS/s 1M point	DS-5514
100MHz 2ch 1GS/s 1M point	DS-5512

• 1M-point long memory & high-speed startup • Made in Japan • Superior cost performance

Features & New Functions

High-speed Startup

Waveforms can be measured a few seconds after switching on the power.

1M point/CH Long Memory

Equipped with a long memory to enable waveforms to be saved over long periods of time and to enable high-quality waveforms that maintain high-speed sampling to be saved.

Fitted with the First Touch Screen of its Class

Perform all operations directly by touching the menu. The touch screen is made of glass with superior levels of durability.



Fitted with Independent Channel Operation Keys and Knobs

Despite its compact size and narrow width, complex function switching has been eradicated to produce easy and comfortable operations.



New Function Added: The new function will be provided in a free upgrade.

Waveform Mask Judgment Function

Pass/fail judgment with the waveform mask.

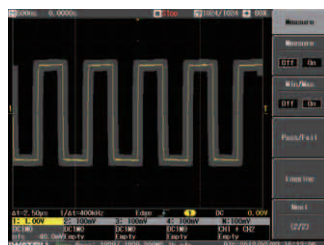
(1) Mask judgment function

The quality of the waveform is evaluated based on the waveform set as the standard. It is possible to display the measurement parameters at the same time as waveform judgment.

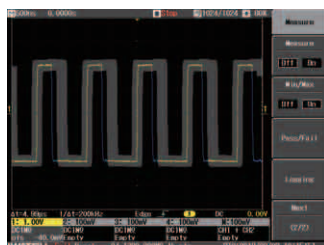
(2) Data Saving Function

The judgment results can be saved onto a USB storage device (in the CSV format with separation by comma.)

If the data is to be saved directly onto a USB storage device, insert the storage device first of all and then save the data onto the USB memory.



Pass Display



Fail Display

Parameter Logging Function and Parameter Judgment Function

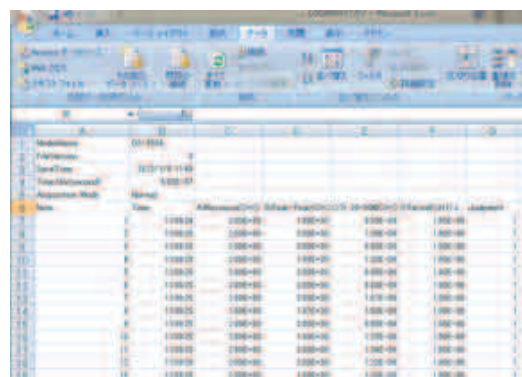
(1) Parameter logging function and parameter judgment function.

Enables the parameter judgment results to be displayed as a list (a list of measurement results only can also be displayed.)



(2) Data Saving Function

The results of measurements can be saved onto a USB storage device (in the CSV format with separation by comma.)



Mask creation function:

Range of 0.09div to 4.00div for the vertical axis and 0.08div to 4.00 for the horizontal axis against the standard waveform.

Target source selection:

CH1, CH2, CH3, CH4 and MATH (calculation). (CH3 and CH4 are available on 4ch models.)

Page Search Function

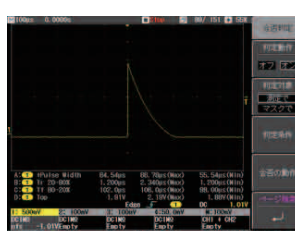
(Searches for the judgment results of measured parameters)



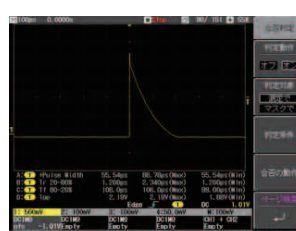
Searches and displays the waveforms (pages) set at [Pass] from the pass/fail judgment results that stipulate the measurement parameter judgment conditions acquired from the waveforms obtained when the RUN/STOP button was pressed to halt waveform reading in the past.

In this example, a [Pass] has been judged because the CH1 positive pulse width is within the range of $60.0\mu s \leq$ and the parameter value $\leq 70.0\mu s$.

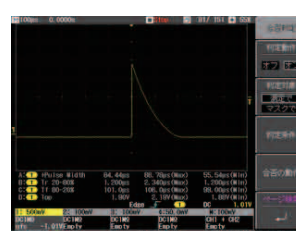
The results obtained from this judgment are shown below.



1) Example of a [Pass] on the 89th item of data from 151 items of data.



2) Example of a [Fail] on the 90th item of data.



3) Example of a [Pass] on the 91st item of data.

DS-5500 Series Special Options

GPIO Interface

DS-576

AUX I/O Option (CH1/CH2 Output)

DS-577

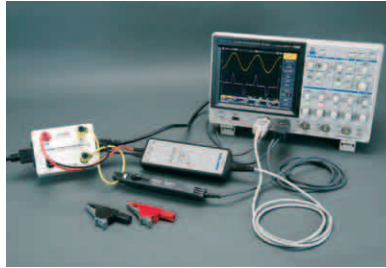
AUX I/O Option (CH1/TRIG Output)

DS-578

AUX I/O Option (Probe Power Supply)

DS-579

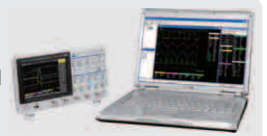
(Can be used as a power source for current probes. Eligible models: SS-240A / SS-250 / SS-260 / SS-270)



Viewer Software

Iwatsu Scope Viewer (free software)

- Enables waveforms to be loaded onto personal computers so that they can be saved as direct value waveform data and image files.
- Advanced cursor measurements can be made on the loaded waveform data within the personal computer.
- Simple remote operations for the oscilloscope can also be made by connecting it with a USB cable.
- Operates on Windows XP and later operating systems.



DS-5500 Series Specifications

Model		DS-5554	DS-5552	DS-5534	DS-5532	DS-5524	DS-5522	DS-5514	DS-5512
Frequency bandwidth(~3dB)		500MHz		350MHz		200MHz		100MHz	
Rise time(Typical)		750ps		1ns		1.75ns		3.5ns	
Input Channel		4	2	4	2	4	2	4	2
Maximum Sampling Speed		1GS/s(all CHs)/ 2GS/s(Channel combined)						1GS/s	
Equivalent Sampling Rate		100GS/s							
Peak detect resolution		1ns							
Averaging		2 to 256 times							
Maximum Memory Length		1Mpts/ch(for all CHs)							
Vertical Resolution		8-bit							
Vertical System	Range	2mV/div - 10V/div(1MΩ),2mV/div - 2V/div(50 Ω)				2mV/div - 10V/div(1MΩ)			
	Offset	2mV/div - 50mV/div : +/-1V,50.2mV/div - 500mV/div : +/-10V,502mV/div - 10V/div : +/-100V							
	DC accuracy	+/- 1.5% + (0.5% of fullscale)							
Maximum Input Voltage		+/-400Vpeak CAT I(1MΩ), 5Vrms(50 Ω)				+/-400Vpeak CAT I(1MΩ)			
BW Limit		20MHz,100MHz				20MHz			
Input coupling		GND, DC 1MΩ, AC 1MΩ, DC 50 Ω				GND, DC 1MΩ, AC 1MΩ			
Input Impedance		1MΩ +/-1.5% // 16pF,50 Ω +/-1.5%				1MΩ +/-1.5% // 20pF			
Probe sense		Automatic 1:1, 10:1, 100:1, 1000:1, Manual 1:1, 5:1, 10:1, 20:1, 50:1, 100:1, 200:1, 500:1, 1000:1, 2000:1							
Probe (One per channel)		SS-101R				SS-0130R			
Timebase		500ps/div - 50s/div		1ns/div - 50s/div		2ns/div - 50s/div		5ns/div - 50s/div	
Roll mode		50ms/div - 50s/div(100kS/s, max.)							
Timebase (Clock) accuracy		10ppm							
Trigger Function		Edge, Edge ALT, Edge OR, Pulse Count, Pulse Width, Period, Dropout, TV, Pattern(OR, NOR, AND, NAND)							
Trigger System	TV mode / Line number /Field sequence	NTSC, PAL, Custom / upto 3,000 / 1, 2, 4 & 8 fields							
	Pulse Count Trigger / Pulse Width Trigger	1 to 9,999events / 15ns to 50s							
	Period Trigger	40ns to 50s							
	Dropout Trigger	50ns to 50s							
	Pattern Trigger	OR, NOR, AND, NAND							
	Trigger source / Status	Input CHs / High , Low , Don't-care							
	Trigger threshold level setting	Individually available for each CH							
Trigger Source		Input CHs, Line, EXT(+/-0.5V), EXT10(+/-5.0V)							
Trigger Coupling		AC, DC, HF Rej, LF Rej, Noise Rej							
Display size/ resolution		7.5inch color TFT-LCD with Touch screen / VGA(640*480pixels)							
Display mode		Y-T, XY, XY (Triggered)							
Vector Display Method		Interpolation on sample points or Dots							
Analog persistence mode		Monochrome grayscale or Color Spectrum							
Persistence time setting		100ms, 200ms, 500ms, 1s, 2s, 5s, 10s & Infinite							
Reference waveform memory		5waveforms							
Panel setting memory		5settings for Internal memory or USB memory							
Parameter measurement		Maximum, Minimum, Peak-Peak, RMS, Cycle RMS, Mean, Cycle Mean, Top, Base, Top-Base, +Overshoot, -Overshoot, Tr 20-80%, Tf 80-20%, Tr 10-90%, Tf 90-10%, Freq., Period, +Pulse Count, -Pulse Count, +Pulse Width, -Pulse Width, Duty Cycle, Integral, Skew (+, -), Skew at level							
Cursor		Time, Amplitude, Time and Amplitude, Value at cursor							
Zoom		Zoom key enable display at individual grid area							
Calculation		Addition, Subtraction, Multiplication, FFT(8k points maximum, RECTANGULAR, HANNING, FLATTOP)							
Rescaling / Unit conversion		a*x+b(x: input voltage at User defined a and b) / Volt, Ampere, Watt, degree and unit-less							
Replay		Automatic waveform recording upto 2,048waveforms, History Replayable							
Counter		6-digit							
Interfaces		USB 2.0(Host&Device), 10/100-BaseTX LAN, IEEE488.2 GPIB(Factory option: DS-576)							
Multi Language Help		Japanese, English, Simplified Chinese							
AUX Interface		AUX connector for External options							
EXT Output(option)		CH1&CH2 output: DS-577, CH1&TRIG output: DS-578							
Waveform Data Storage		USB memory for Binary, ASCII, Mathcad, Calculation(ASCII) & Calculation(Mathcad)							
Hardcopy Output		Output to USB memory in TIFF, BMP & PNG format or Output to PictBridge Printers							
AC input voltage, Frequency, Power consumption		90 to 264V AC at 50Hz/60Hz, 90 to 132V AC at 50Hz/60Hz/400Hz, 95VA (60W) max.							
Dimensions / Weight		330(W)*190(H)*124(L) mm / 3.7kg							
Performance guaranteed Temperature range		+10℃ to +35℃							
Operation temperature and humidity / Altitude conditions		0° to +40℃ at 5 to 80%(RH≦30℃) and 55%RH or less at 40℃ non-condensation / 2,000meter or lower							
Storage temperature		-20℃ to +60℃							

New function

Pass/Fail	Judgment mode	During parameter judgment	During mask judgment
	Judgment target:	select one of four parameter types (A, B, C, D)	select one among CH1 to CH4 (2CH machine: CH1 to CH2) and Math
	Comparison ranges:	Large/small comparison, within range/outside range comparison	All In or All Out for 1 to 4 areas
	Mask creation:	—	Horizontal or vertical direction, from among CH1 to CH4 (2CH machine: CH1 to CH2), Math, and REF1 to 5
Logging	Actions during Pass/Fail judgment	Stop waveform capture/Save screen (USB memory)/Save waveform (USB memory)/Beep/Pulse output (when equipped with DS-578 option)	
	Logging items	Time and parameter measurement results (conditions A, B, C, D), Pass/Fail judgment results	
	Log output destinations	When recording : Pop-up window, internal memory (up to 86400 entries)	
		After recording : USB memory	

Analog Oscilloscope SS-7800A Series

Familiar operation response.
Origin of waveform
measurements.
Limitless gradation display.
Eternal standard unit.

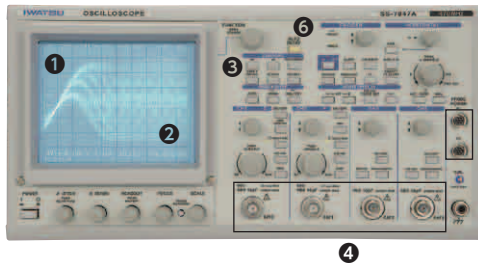


DC - 470MHz 4ch Oscilloscope
SS-7847A
DC - 400MHz 4ch Oscilloscope
SS-7840A
DC - 300MHz 4ch Oscilloscope
SS-7830A
Programmable Oscilloscope
SS-7840AP /
SS-7830AP



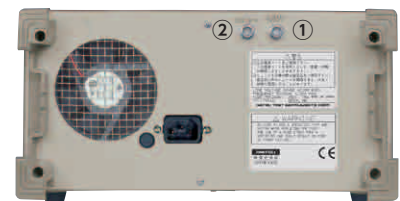
The world's only analog wideband oscilloscope

- DC offset equivalent to a maximum of ± 500 div
- Built-in 6-digit counter that counts frequencies up to 400MHz
- Optimal high-speed auto setup functions at the press of a key for the vertical axis, time axis and trigger level
- Save/recall functions for setting up a maximum of 256 panels setup
- Programmable setup type, SS-7840AP and SS-7830AP
- Equipped with a bright and sharp mesh-less CRT



- 1 High-brightness CRT (Japan made)**
High-accuracy 6-inch, meshless CRT with internal graticule displays waveforms with bright and sharp traces.
- 2 High-accuracy 6-digit frequency counter**
A frequency counter with $\pm 0.0025\%$ accuracy is built in.
- 3 Save/recall function**
Up to 256 different setups with 12-character comments can be saved and recalled.

- 4 Wide frequency bandwidth of DC - 470MHz (SS-7847A)/400MHz (SS-7840A)/300MHz (SS-7830A)**
CH1/CH2 sensitivity: 2mV/div ; CH3/CH4: 100 mV/div or 500mV/div selectable ~ 5 V/div
- 5 Power supply output terminals for FET or current probes**
Optional SFP-5A/4A (DC - 1GHz/800MHz) FET probe and SS-250 (100MHz)/SS-240A (50MHz) SS-260/SS-270 (High current) current probes can be used.
- 6 Quick auto setup**
Automatically displays an optimum range for input waveform (CH1/CH2)



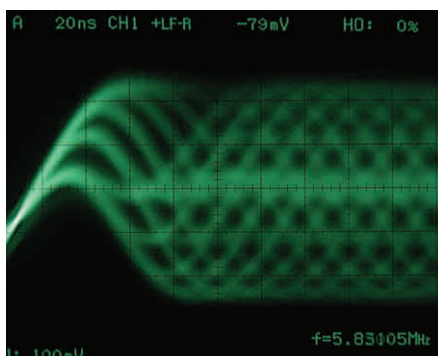
- 1 External intensity modulation signal input**
0.5Vp-p, DC - 5MHz, ± 40 V
- 2 CH2 signal output**
20mV/div, DC - 200MHz/50 Ω

Indispensable for a wide range of requirements

Eye-patterns in optical disc manufacturing process

When evaluating optical discs such as Blu-ray Discs, HD-DVDs, CDs or DVDs, eye patterns need to be observed. With this analog oscilloscope, accurate observation of the eye patterns of highspeed and high-density media is easily possible.

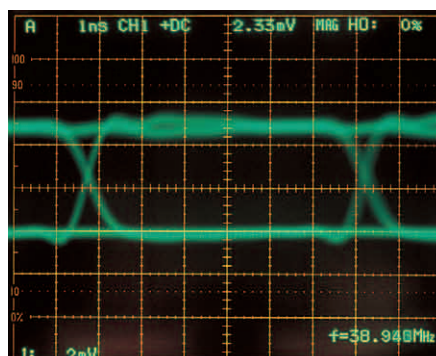
* Blu-ray Disc signal eye pattern waveform



ATM 155Mbps signal eye patterns

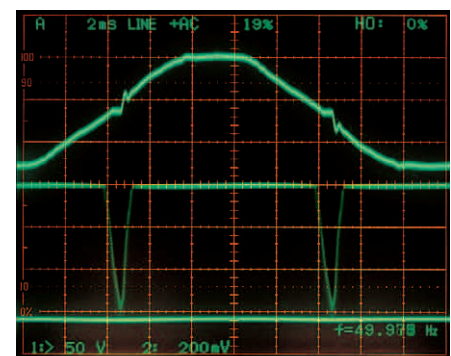
The standard transmission rate for most networked communication systems is 155 Mbps (STM-1). The amount of jitter can be estimated by observing the signal waveform with the eye pattern and following the pulse mask standard.

* 155 Mbps signal eye pattern waveform measured with SS-7847A (DC - 470MHz)



Switching power supply measurement

A switching power supply unit with a higher harmonics measure switches the voltage of a commercial power supply at high speed. In terms of circuit operation, switching stops at the zero cross of the AC power supply. To observe this condition, an analog oscilloscope is required. Analog oscilloscopes are also superior when simultaneously observing voltage and current waveforms. In addition, when magnifying a switching waveform for observation on an analog oscilloscope, no complicated operations are required to trigger the waveform.



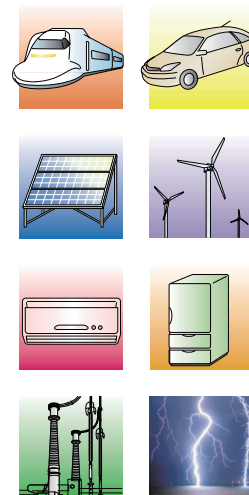
SS-7800A Series Specifications

Model				SS-7847A		SS-7840A		SS-7830A		
Display	CRT			CRT 6-inch rectangular, internal graticule (8 x 10 div, with scale illumination) - Japan made						
	Accelerating voltage			Approx. 20kV						
Vertical deflection system	Mode			CH1, CH2, CH3, CH4, ADD, ALT, CHOP						
		Frequency bandwidth	2mV/div -5V/div	—		DC - 400MHz		DC - 300MHz		
			5mV/div -50mV/div	DC - 470MHz		—		—		
			2mV, 100mV/div -5V/div	DC - 440MHz		—		—		
	VSWR				Less than 1.35 through DC - 400MHz(with 50 Ω input)				Less than 1.35 through DC - 300MHz (with 50 Ω input)	
		Sensitivity(CH1, CH2)			2mV/div - 5 V/div 11-step (1-2-5), 2mV - 12.5V/div (with vernier)					
		Accuracy			± 2%					
	Bandwidth limiter			20MHz or 100MHz				20MHz		
	Rise time(Calculated from freq. bandwidth x rise time =0.35)			745ps at 20mV		875ps		1.17ns		
	Signal delay			Available						
	Input coupling			AC, DC, GND(The cut-off freq. is 4Hz with AC coupl.)						
	Input RC: direct			1MΩ input :1MΩ ±1.5% // 16pF ±2pF / 50 Ω input : 50 Ω ±1%						
	Max. input			1MΩ input : ±400Vm max. / 50 Ω input : 5V RMS						
	Probe sensor			1:1 , 1:10, 1:100 detection possible						
	Sensivity (CH3, CH4)			100mV/div, 500mV/div ± 3% (1MΩinput)						
	Input coupling (CH3, CH4)			AC, DC						
Offset voltage variable range			Offset voltage: variable axis range ± 1V : 2mV/div - 50mV/div ± 10V: 0.1mV/div -0.5 mV/div ± 100 V: 1V/div - 5V/div							
Triggering	A triggering	Source	CH1, CH2, CH3, CH4, LINE							
			AC, DC, HF-REJ, LF-REJ							
		Level	DC-10MHz	0.4 div						
			10MHz - 100MHz	1.0 div						
			100MHz - 300MHz	—				2.0 div		
	100MHz - 400MHz		2.0 div				—			
	B triggering	Source	CH1, CH2, CH3, CH4							
		Coupling	AC, DC, HF-REJ, LF-REJ							
		TV triggering	NTSC, PAL/SECAM, HDTV							
		Trigger mode	TV-V (ODD, EVEN, BOTH) line number selectable, TV-H							
TV clamp		Back porch level								
Event trigger	Count	Setting range: 1 - 65535, Max. count freq. : 50MHz								
	Burst	Time setting range: 0.1μs - 99.9s								
X-Y	X-axis			CH1						
		Sensitivity		Same as CH1						
		Accuracy		± 2%						
		Bandwidth		DC -2MHz						
	Y-axis	Operating channel		CH1, CH2, CH3, CH4, ADD						
Horizontal deflection system	Horizontal display			A, ALT, B, X-Y						
	Mode			AUTO, NORM, SINGLE						
	A sweep time	Fastest range with mag.		500psec/div			1nsec/div			
		Slowest range		500msec/div						
	B sweep time	Fastest range with mag.		500psec/div			1nsec/div			
		Slowest range		20msec/div						
		Accuracy A/B		± 2%						
		Delay		Triggered delay: CH1 , CH2, CH3, CH4 Continuous delay: RUNS AFTER						
	Delay jitters			1/20,000						
Magnifier			10 times							
Hold-off time			Variable (up to sweep length or longer)							
Auto setup				V/H position, V/H range, trigger						
	Input channel			CH1, CH2						
	Frequency response			50Hz -100MHz						
CH2 signal output (50Ω)				DC - 200MHz			DC -100MHz			
Z axis input				0.5Vp-p						
Calibrator output				Square-wave, 1kHz ±0.1%, 0.6Vp-p ± 1%						
Power supplv for probe				Voltage: ± 12V, 2 FET or current probes connectable, Offset control available						
CRT read-out	Read-out			Attenuator range, ADD UNCAL, AC/DC/GND, vertical mode, CH2 polarity A/B sweep range, x10 MAG, UNCAL, horizontal display mode, hold-off trigger coupling, trigger source, trigger slope, TV-field, TV-line, TV-system						
	Cursor measurement			ΔV(voltage measurement), Δt(time measurement), 1/Δt(division calculation by Δt)						
Frequency counter				6-digit, ± 0.0025%						
	Frequency range			2Hz - 400MHz , reciprocal						
Save and recall				Up to 256-setup						
	Power off set-up			Panel setting before switch is powered off						
	Back-up			Battery back-up (approx. 30,000 hrs)						
Power supply				AC 100V - 240V, 50/60Hz, max.130VA						
Dimensions and weight				320W x 160H x 420L (mm), approx. 8.5kg						
Environment conditions		Operating		0°C - +40°C, 90% RH (40°C)						
		Performance guaranteed		+10°C - +35°C						
Standard accessories				Power cable (1), probe (2), fuse (2), panel cover (1), operation manual (1), accessory bag (1)						

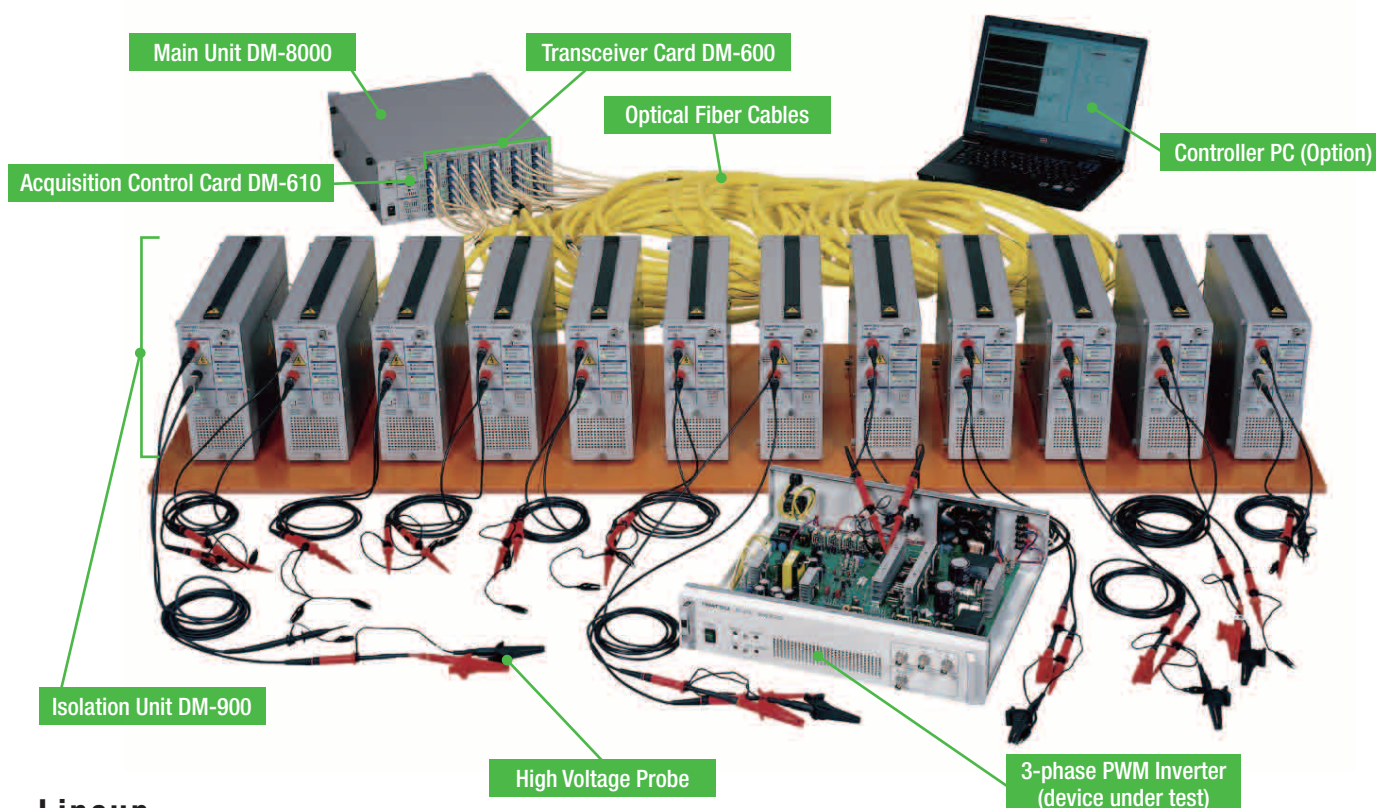
Isolation measurement system

DM-8000

- The input block, control block and display block are isolated with optical fiber cables. (DM-900/L, DM-910/L)
- Frequency bandwidth: DC to 500MHz.
- Simultaneous multi-channel measurement of many channels of different reference potentials. (2 to 24 channels) (DM-900/L, DM-400/L)
- Long-life battery drive. (The system can be driven by three batteries for about 12 hours) (DM-900/L, DM-910/L)
- Measurement using long memory. (DM-900/L, DM-910/L, DM-400/L)
- Simultaneous measurements of the inverter's switching waveform and ON-voltage. (DM-910/L)
- Also supports synchronous measurements with the non-isolated unit. (DM-400/L)



Example: Isolation Unit DM-900 x 12units (24ch)



Lineup

Items	Model
Main unit	DM-8000
High-speed main unit *1	DM-8000H
Acquisition control card	DM-610
Transceiver card (optical x 2)	DM-600
Transceiver card (optical x 1, metal x 1)	DM-620
Transceiver card (metal x 2)	DM-630
Isolation unit (500k points) *2	DM-900
Isolation unit (16M points) *2	DM-900L
Isolation unit (high resolution, 500 k points) *3	DM-910
Isolation unit (high resolution, 16 M points) *3	DM-910L
Acquisition unit (500k points) *4	DM-400
Acquisition unit (16M points) *4	DM-400L

*1 Performance is improved when using three or more units with a memory length exceeding 100k points.

*2 With insulation case

*3 With insulation case. Optional probe is required for voltage measurements.

Items	Model
Optical fiber cable S (2m) *5	DM-002
Optical fiber cable S (5m) *5	DM-004
Optical fiber cable (5m)	DM-005
Optical fiber cable (10m)	DM-006
Optical fiber cable (20m)	DM-007
Optical fiber cable (50m)	DM-008
Optical fiber cable (100m)	DM-009
Optical fiber cable (200m)	DM-010
Acquisition cable (2m)	DM-105
Acquisition cable (5m)	DM-106
Battery pack	DM-551
Battery pack (set of three battery packs) *6	DM-553

*4 Non-isolation type unit driven by AC power only.

*5 Optical cable set without sheath.

*6 Standard item for isolation unit.

Isolation with Optical Fiber cable (2 to 200 m)

The input block, control block and display block are isolated by an optical fiber cable. Owing to the fact that isolation units are isolated from each other by optical fiber cables, it is possible to simultaneously measure signals that have different reference potentials, such as signals from the high and low-side switch of an inverter or from the primary and secondary sides of a power converter.

DM-8000 main unit / High-speed DM-8000H main unit

Up to 12 isolation units and acquisition units can be connected. An acquisition control card for capture control and up to 6 specially designed transceiver cards can be installed onto the main unit. The gigabit Ethernet-enabled high-speed main unit (DM-8000) improves the waveform update speed when using 3 or more units. The interlock control terminal is on the rear panel.

DM-600 transceiver card

Two isolation units can be connected per card.

DM-620 optical and metal transceiver card

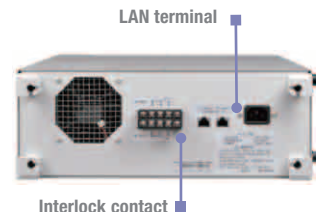
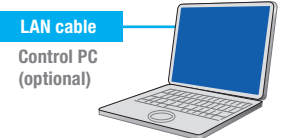
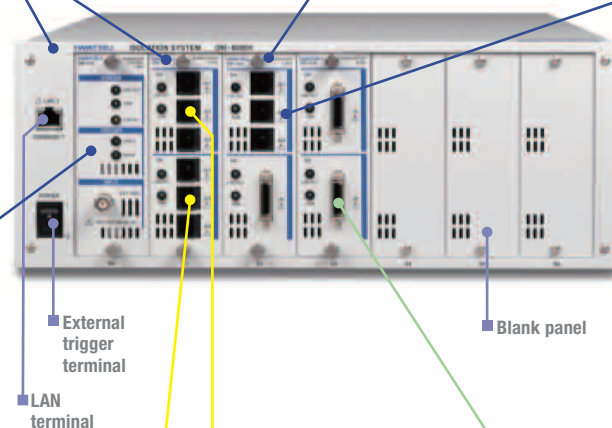
One isolation unit and one acquisition unit can be connected per card.

DM-630 metal transceiver card

Two acquisition units can be connected per card.

DM-610 acquisition control card

This card controls waveform capture in measurement units. It also provides a non-isolated external trigger input, which can be changed to an external trigger output terminal.



Optical fiber cable

Optical fiber cable

Acquisition cable

DM-900 (500k)/DM-900L (16M) isolation units



The units are operated by a built-in battery to perform floating measurements.

Frequency bandwidth: DC to 500MHz, highest sampling rate: 2GS/s, memory length: 500k points (DM-900), 16M points (DM-900L), input: 2channels (not isolated), interface: optical interface (set of three interfaces)



Insulation case

Withstand voltage: 10kV (Standard accessory)

DM-910 (500k)/DM-910L (16M) isolation units (high resolution)



The units are operated by a built-in battery to perform floating measurements.

The high resolution enables the simultaneous measurement of switching waveforms and on-voltage. Frequency bandwidth: DC to 500MHz, highest sampling rate: 2GS/s, memory length: 500k points (DM-910), 16M points (DM-910L), input: 1channel, interface: optical interface (set of three interfaces)



Insulation case

Withstand voltage: 10kV (Standard accessory)

DM-400 (500k)/DM-400L (16M) acquisition units



The units can continuously operated with an AC power source.

These units are best suited to the non-isolated measurement

of grounded power probes, for example. Frequency bandwidth: DC to 500MHz, highest sampling rate: 2GS/s, memory length: 500k points (DM-400), 16M points (DM-400L), input: 2channels (not isolated), interface: electric interface (one set)

DM-553 Li-ion battery (built-in)

The battery can be inserted or removed from the front of the isolation unit. It uses three batteries to enable the unit to continuously operate for 20hours. The battery can be charged with the use of the main unit.



The DM-900/L and DM-910/L are supplied with three batteries.

DM-002 to DM-010 optical fiber cables

The optical fiber cables are resistant to bending and external pressure.



Cable length: 2m to 200m
*1-2-5 step length
Without cover: 2m or 5m
With cover: 5m to 200m

IE-1196 probe especially designed for the DM-910 (custom mode)

The probe is a low distortion probe that supports the DM-910's high resolution.



Frequency bandwidth: DC 200MHz
Probe ratio: 100:1
Cable length: 1.5m

DM-105/DM-106 acquisition cables

Interface cables especially designed for the acquisition unit. These cables are connected to the unit and transceiver by electrical signals from the DM-400/L.



Cable length: 2 or 5m

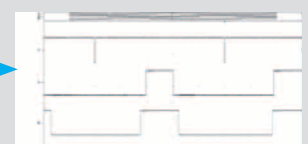
The DM-9xxL long memory isolation unit enables detailed analysis during a basic inverter duty cycle.

The DM-900L and DM-910L long memory isolation units enable detailed analysis of individual carrier signals while capturing a base duty cycle.



Gate driving waveform of the U, V, and W phases on the high side of a 3-phase inverter.

A fundamental duty cycle (16ms on the sample screen) can be captured at a rate of 1GS/s.

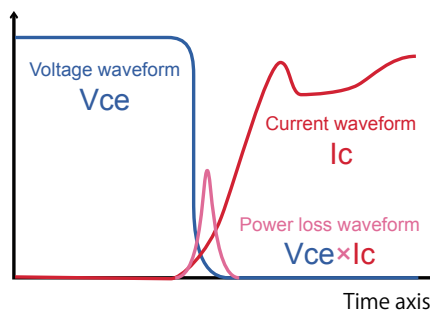


View with zoom display.

Up to 24 CH at a high voltage and wide bandwidth can be simultaneously measured.

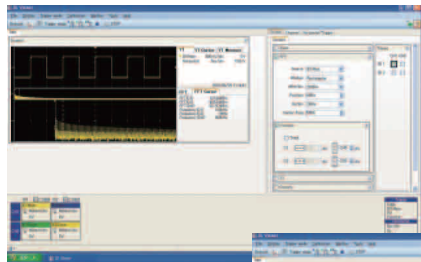
Waveform monitoring and other system operations are remotely performed using the standard IS Viewer (software). The IS Viewer can be used off-line as well, and is therefore useful for data organization at locations remote from the measurement site.

The many operation functions provided by the IS Viewer facilitate power loss and other measurement.

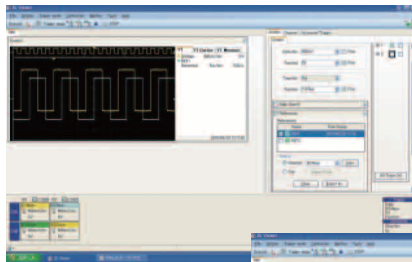


The Vce, Ic, output voltage and current waveforms of the upper and lower arms of an inverter can be simultaneously measured. dv/dt, di/dt, and other parameters, such as power loss, can be easily calculated from the measurement waveforms.

Functions of the IS viewer (DM-800)



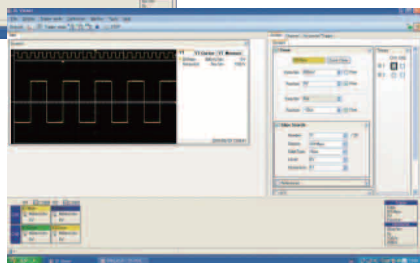
FFT function
This function is used for the frequency analysis of measured waveforms.



Reference display function
This function is used to compare waveforms.

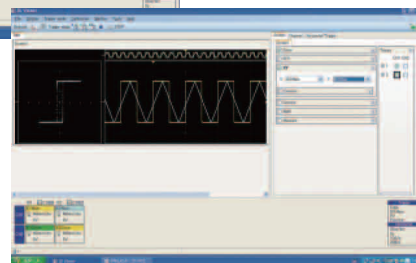
Edge search function

This function is used to automatically detect the edge of a monitored waveform and display selected edges.

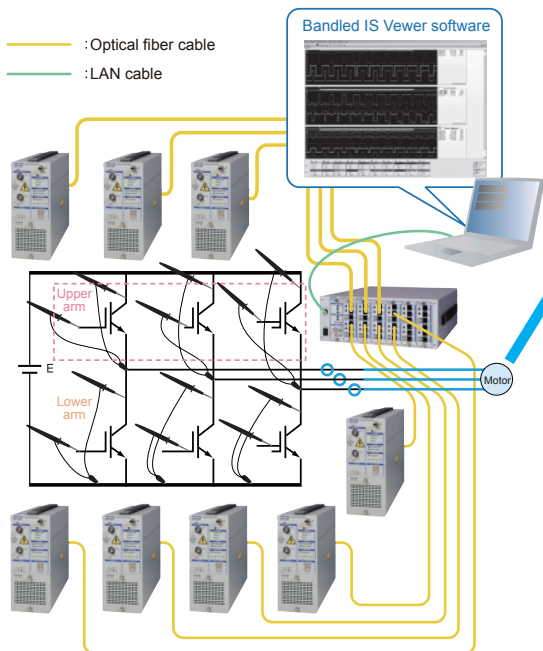


X-Y display function

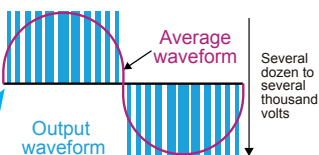
This function is used to evaluate the SOA (safe operation area) and other items.



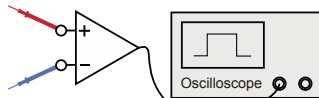
Multi-channel floating measurements (simultaneous measurement example of the upper and lower arms of a 3-phase inverter)



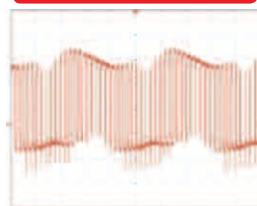
The waveform of voltage output from the 3-phase inverter that drives a motor or other device (shown in the left-hand figure) is a pulse voltage waveform, as shown in the figure below.



Measuring Vge of the upper arm with differential input

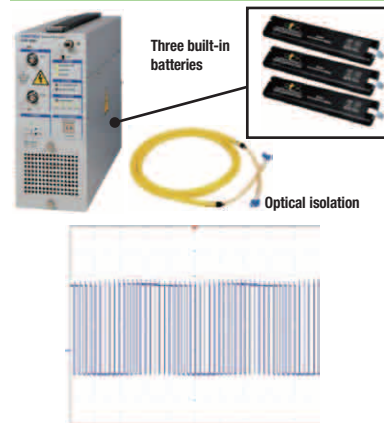


The common mode noise prevents accurate measurement.

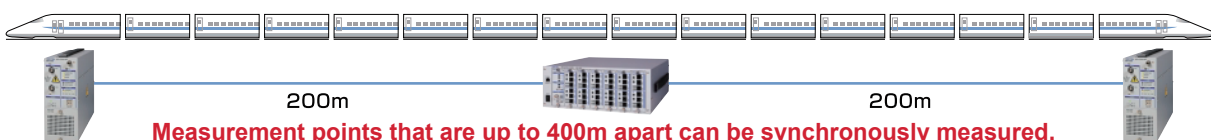


Differential probes were used for this type of measurement in the past, but this resulted in the waveform sometimes being distorted, and it was sometimes difficult to ensure sufficient measurement bandwidth due to constraints of the common mode rejection ratio or withstand common mode voltage. With optical fiber isolation, this isolation system can accurately monitor signals without being affected by these constraints.

Measuring Vge of the upper arm with isolation input

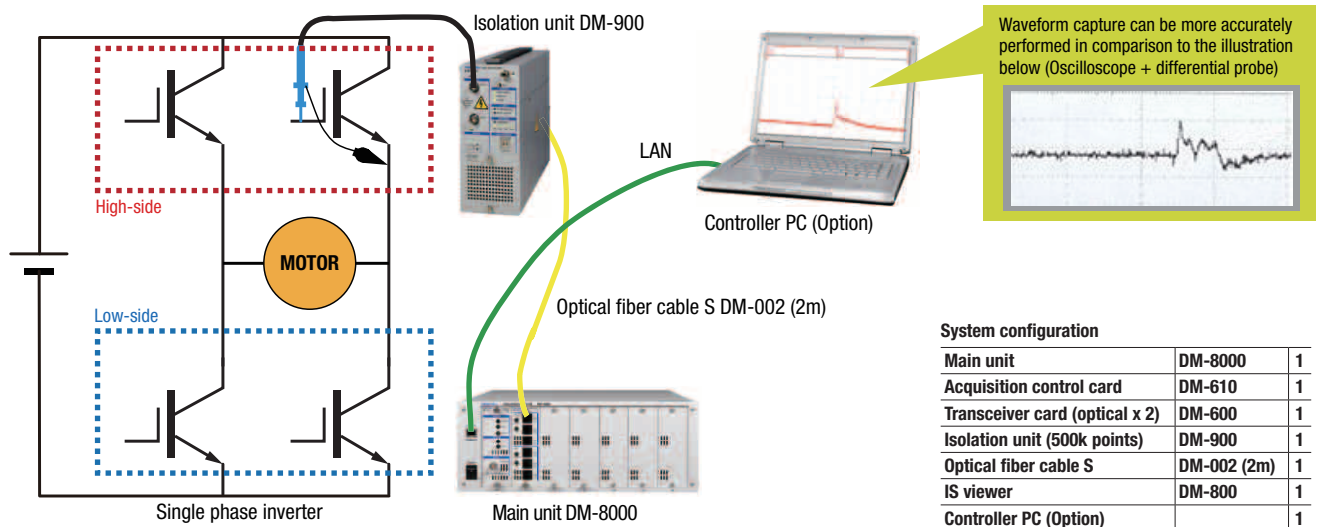


Remote measurement (up to 400m)

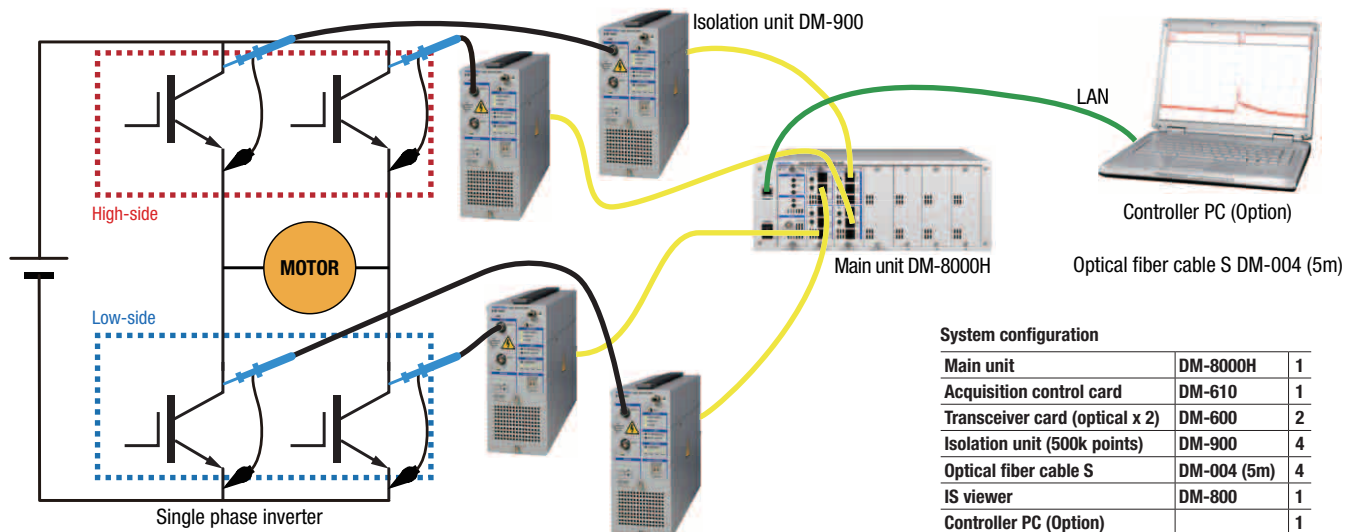


Measurement points that are up to 400m apart can be synchronously measured.

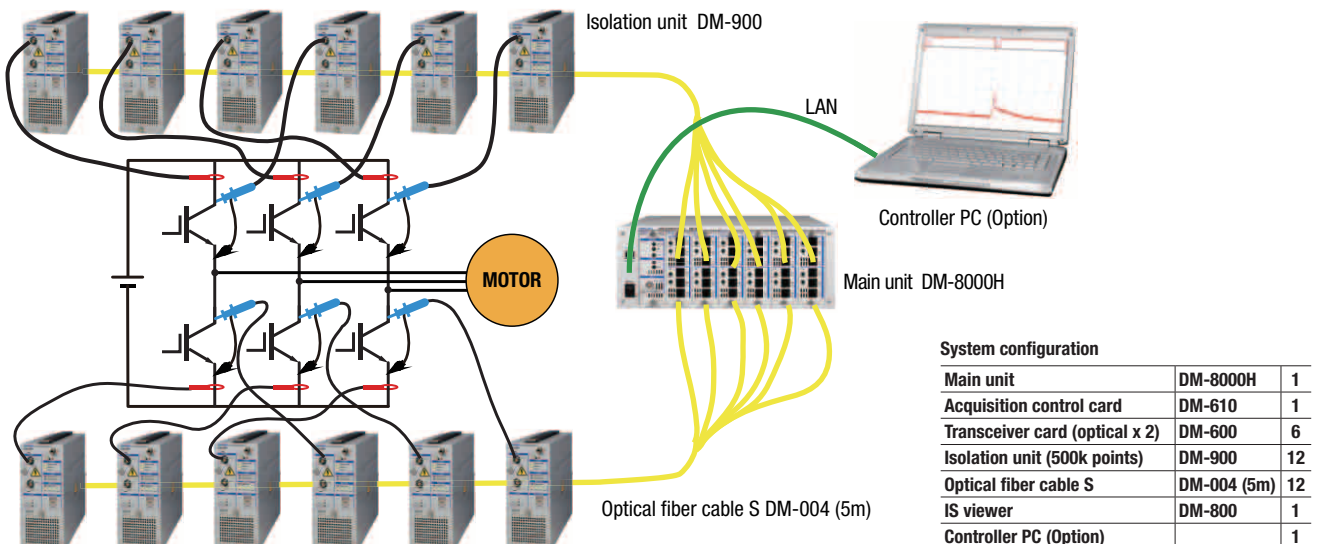
IGBT Gate voltage measurements in the high-side switch of a single phase inverter (one unit)



IGBT Vce voltage measurements in the high-side switch of single phase inverters (four units)



IGBT Vce voltage & Ic current measurements of 3-phase, 2-level inverters (twelve units)



Isolation System DM-8000 Specifications

DM-900L/DM-910L Isolation Unit and DM-400/L Acquisition Unit

Model	DM-900	DM-900L	DM-910	DM-910L	DM-400	DM-400L
Signal input unit						
Frequency Bandwidth (-3 dB)	500MHz					
Bandwidth limiter	20MHz / 100MHz					
Input impedance	1MΩ // 16pF				1MΩ // 16pF or 50 Ω	
Maximum input voltage	400V max (DC+peakAC<=5kHz) CAT I					
Number of channels	2 (between channels are not isolated)		1		2 (Not isolated)	
Input coupling	GND, DC1MΩ, AC1MΩ		GND, DC1MΩ		GND, DC1MΩ, AC1MΩ, DC50 Ω	
Input sensitivity	2mV/div~10V/div, 1-2-5 steps		CH1-MAIN: 50mV/div~5V/div, 1-2-5 steps CH2-ZOOM: 2mV/div~1V/div, 1-2-5 steps		2mV/div~10V/div, 1-2-5 steps	
Offset range	2mV/div~50mV/div, ± 1V * ¹ 100mV/div~500mV/div, ± 10V * ² 1V/div~10V/div, ± 100V * ³		CH1-MAIN: 50mV/div~500mV/div, ± 10V * ² 1V/div~5V/div, ± 100V * ³ CH2-ZOOM: 2mV/div~20mV/div, ± 2V * ¹ 50mV/div~1V/div, ± 20V * ²		2mV/div~50mV/div, ± 1V * ¹ 100mV/div~500mV/div, ± 10V * ² 1V/div~10V/div, ± 100V * ³	
Offset accuracy	± (1.0% + 0.5% of full-scale + X) X: * ¹ 1mV, * ² 10mV, * ³ 100mV					
DC gain accuracy	± (1.5% + 0.5% of full-scale)					
Probe sensitivity	10:1, 100:1, 1000:1 (Auto detection/manual settings)					
Sample rate	1GS/s (2GS/s during interleave)					
Vertical axis resolution	8bits					
Maximum memory length	500k points/ch	16M points/ch	500k points/ch	16M points/ch	500k points/ch	16M points/ch
Trigger system unit						
Trigger sources	CH1, CH2		CH-1-MAIN		CH1, CH2	
Trigger slope	Positive / Negative					
Coupling	AC, DC, HFREJ, LFREJ					
Level range	125% of full-scale					
Interface						
Interface	1 set of 3 optical interfaces (optical fiber cable: 2m to 200m)				1 set of electrical interfaces (wire cable: 2 or 5m)	
Power supply and battery unit						
Internal battery	3 battery packs (unit can operate on one battery)				—	
Battery charging	Can be charged by the main unit				—	
Power consumption	120VAm _{ax} (when using AC power)				40VAm _{ax}	
Battery operation time	Approx. 12 hours of continuous operation (when using 3 batteries)				—	
Battery charging time	Approx. 6 hours				—	
AC power supply	AC100 to 240 (50/60Hz)					
Calibration signal						
Calibration signal	0.6V / 6V (selectable)					
Mechanical unit						
Dimensions (mm)	122.4 (H) X 258.4 (W) X 544 (D)				96.4 (H) X 171.6 (W) X 322.6 (D)	
Weight	Approx. 7kg (excluding battery packs and accessories) Battery pack weight: Approx. 660g per pack				2.6kg	
Operating temperature	0℃ to +40℃					
Performance guaranteed temperature	+10℃ to +35℃					
Accessories						
Battery pack	3				—	
Power supply cable	1					

DM-8000/DM-8000H Main Unit

* When the DM-610 acquisition control card is installed

Transceiver card connection	
Number of slots	6 (Max. 12 isolation units and/or acquisition units can be connected.)
Time axis	
Sweep range	1ns/div to 20s/div
Clock accuracy	10ppm
Acquisition mode	Normal, peak
Trigger system	
Mode	Auto, Normal, Single, Stop
Source	Up to 24 CH
Type	Edge, Pulse width
Trigger delay	Available
Interface	
Ethernet port	DM-8000: (10BASE-T/100BASE-TX)×3, DM-8000H: (1000BASE-T)×3
Power supply unit	
AC power supply	100V to 240V (50/60Hz)
Power consumption	DM-8000: 100VA max, DM-8000H: 130VA max
Mechanical unit	
Dimensions and weight	132(H) × 351(W) × 420(D), Available. 6.9kg
Operating temperature	0°C to +40°C
Performance guaranteed temperature	+10°C to +35°C
Accessories	
LAN cable	1
Power supply cable	1
Operation manual	CD-R(1)
Control software	IS Viewer DM-800 CD-R (1)

Note #1: Intel and Pentium are registered trademarks or trademarks of Intel Corporation and its subsidiary companies in the United States of America and other countries.

Note #2: Windows is a registered trademark or trademark of Microsoft Corporation in the United States of America and other countries.

DM-600/DM-620/DM-630 Transceiver Card

Number of isolation / acquisition units connected	DM-600: 2 (DM-900/L, DM-910/L)
	DM-620: 1 (DM-900/L, DM-910/L) +1 (DM-400/L)
	DM-630: 2 (DM-400/L)
Operation indicator	Status display via LED
Mechanism	Card inserted in main unit (DM-8000)
Operating temperature	0°C to +40°C
Performance guaranteed temperature	+10°C to +35°C

IS Viewer DM-800 (supplied with the DM-8000/DM-8000H main unit)

* IS Viewer is installed in the controlling computer (option) and is used to operate the isolation system and to monitor waveforms.

Main function	
Operations	+, -, ×, ÷, × , ÷ , f, dy/dx
Parameter measurements	Max, Min, p-p, Top, Base, Top-Base, RMS, Cycle RMS, Mean, Cycle Mean, +/- Overshoot, Transition Time, dv/dt, Freq, Period, +/- Pulse Count, +/- Pulse Width, Duty, Integral, Integral (abs), Integral (pos), Integral (neg), Skew (%), Skew (Level)
Other functions	XY display, FFT, Cursor, smoothing, channel de-skew, re-scale, off-line viewer
Waveform storage	CSV
Saving images	BMP, PNG, Clipboard
Saving setups	with / without waveforms
Controlling computer	
CPU	Intel® Pentium® 4 Processor or later ^{Note #1}
RAM	2GB or larger
OS	Windows® XP Professional SP3 ^{Note #2} Windows® Vista Business SP2 ^{Note #2}
Display	At least WXGA (1,280 x 768 pixels) recommended (SXGA (1,280 x 1,024 pixels) is required for full display.)

XY Monitor and Multi-Window Waveform Monitor 8ch Model

SV-1014 Series

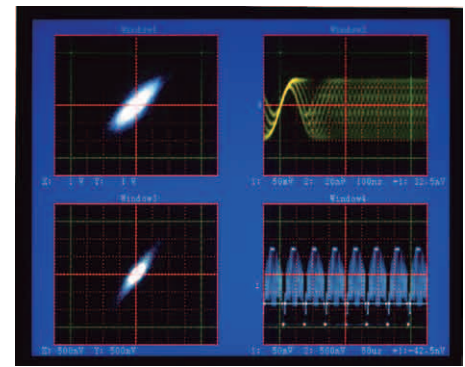
Custom Order

* Standard specifications require operation control to be performed with an application installed onto a PC. The operation panel is optional.

Compact,
power-saving,
multi-channel,
high-speed waveform updates

Optimized by enabling more than one
oscilloscope to be used simultaneously and
incorporating adjustment lines and functions.

- Compact, energy-saving specifications (power consumption: 30VA or less) achieved through digital processing (one-tenth of our conventional models)
- Multi-channel and multi-monitor functions (the signals for two independent channels can be displayed on a maximum of four windows)
- High-speed waveform updates (maximum of 100,000 times per second)
- Gradation displays similar to analog oscilloscopes



Example of the 8CH model monitor display

* The LCD monitor is equipped separately.

* Contact us for the specifications of the operation panel (option.)

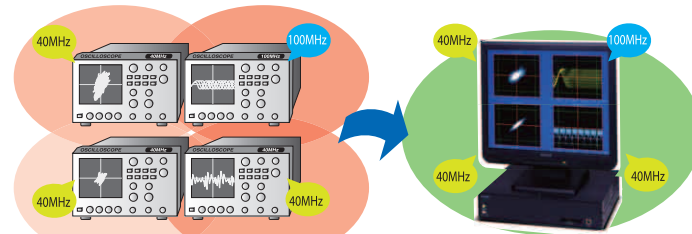


2CH Model with built-in type LCD

* Contact us for the specifications of the operation panel (option.)

Users Opinions

It is very difficult to keep an eye on four different oscilloscope screens while making the necessary adjustments. This unit focuses the line of sight and makes work much more efficient.



The compact shape makes function incorporation a simple task, and the USB control shortens setup time.

Specifications and Performance (Example)

		Window 1 ^{*1}	Window 2	Window 3 (Specially for XY)	Window 4 ^{*1}
Vertical Axis (Y Axis)	Frequency Band (-3dB)	40MHz	100MHz	40MHz	40MHz
	Input Channel	CH1、CH2			
	Input Voltage Range	10mV/div to 5V/div			
	Accuracy	± 2% ± 1dot			
	Input Coupler	AC、DC			
	Input Impedance	1MΩ ± 2% // 20pF ± 2pF			
	Maximum Input Voltage	± 40V(DC + Acpeak)			
Display Area	Display Method	Y-T/X-Y switching possible			
Horizontal Axis (Z Axis)	Sweep Time (1-2-5 steps)	1us/div to 500ms/div	50ns/div to 500ms/div	1us/div to 500ms/div	
Synchronization	Signal Source	CH1 / CH2			
	Slope/Coupling/Synchronization Method	+ / DC / Edge			
	Trigger Level Range	± 10div			
	Minimum Synchronization Sensitivity	1div			
Acquisition	ADC	25MSa/s 8bits			
Output Area	Shape (Display Resolution)	Analog RGB Output (480 dot x 640 dot)			
	Window Display	200dot × 250dot(8div × 10div)			
	Marker Display	2 Vertical, 2 Horizontal Displayed Simultaneously			
Interface		USB1.1			
Power Supply	Input Voltage/Frequency	AC100-240V 50/60Hz			
	Maximum Electrical Power	30VA or less			
Weight, Size	Weight	Approximately 1.8kg			
	Size	Approximately 298W x 82.4H x 320L mm (excluding protrusions)			
Environmental Conditions	Guaranteed Performance Temperature	+23°C ± 5°C			
Option		Operation Panel			

*1: The specifications for the 2CH model with built-in LCD (1 window) can be selected from Window 1 and Window 4.

Passive Probes (100:1, 10:1, 1:1)

Model	Attenuation	Cable length	Input RC	Bandwidth	Oscilloscope input capacity	Conforming models
SS-101R C €	10:1	Approx. 1.2 m	10MΩ, approx. 12pF	DC to 500MHz	13pF to 23pF	DS-5532/5534/5552/5554, DS-5332/5334/5352/5354, TS-8500/P, SS-7840/A/H/P, SS-7830, SS-7825/P, TS-81000/80600, SS-7847A
SS-103R C €	10:1	Approx. 1.2 m	10MΩ, approx. 15pF	DC to 200MHz, ±3dB	16pF to 24pF	SS-7821/A
SS-0110 C €	10:1	Approx. 1.5 m	10MΩ, approx. 22pF ± 2pF	DC to 60MHz, ±3dB	20pF to 45pF	SS-7804/P/7802, DS-8706/8701
	1:1		1MΩ, 200pF or less	DC to 6MHz, ±3dB	—	
SS-0120 C €	10:1	Approx. 1.5 m	10MΩ, approx. 14pF	DC to 100 MHz, ±2 dB	18pF to 35pF	DS-8606/C, DS-8608A/8607A, DS-8608/8607, DS-8710, SS-6123A/6613, SS-6611/6122A
	1:1		1MΩ, 150pF or less	DC to 6MHz, ±3dB	—	
SS-0121 C €	10:1	Approx. 2 m	10MΩ, approx. 16pF	DC to 100MHz, ±3dB	18pF to 35pF	SS-3510, SL-232
	1:1		1MΩ, 170pF or less	DC to 3MHz, ±3dB	—	
SS-0122 C €	10:1	Approx. 1.5 m	10MΩ, approx. 14pF	DC to 100MHz	10pF to 32pF	DS-5110/DS-5106/DS-5110A/DS-5106A DS-5110B/DS-5106B
	1:1		1MΩ, 150pF or less	DC to 6MHz	—	
SS-0112 C €	10:1	Approx. 1.5 m	10MΩ, approx. 22pF	DC to 60MHz	10pF to 45pF	DS-5104/DS-5102/DS-5104A/DS-5102A DS-5104B/DS-5102B
	1:1		1MΩ, 200pF or less	DC to 6MHz	—	
SS-0130R C €	10:1	Approx. 1.5 m	10MΩ, approx. 12.5pF ± 2pF	DC to 100MHz, ±2dB	18pF to 35pF	DS-5512/5514/5522/5524, DS-5312/5314/5322/5324, SS-7811/A/7810/A/7806, DS-8617, DS-9121/9122, DS-8812/8814
SS-0004	1:1	Approx. 1.1 m	44pF ± 6pF	DC to 30MHz, ±3dB (For probes only)		All models with BNC for input
Maximum input voltage						
SS-0160R C €	100:1	Approx. 2 m	66.7MΩ ± 1 %/3pF or less	DC to 400 MHz	6pF to 18pF	6,000 V CAT I (DC+ACpeak) 1,000 V CAT II (DC+ACpeak)
SS-0161R C €	100:1	Approx. 2 m	66.7MΩ ± 1 %/3pF or less	DC to 400 MHz	6pF to 18pF	4,000 V CAT I (DC+ACpeak) 1,000 V CAT II (DC+ACpeak)

- Connector: BNC
- Maximum input voltage: 600 V (DC + AC peak); 10: 1 probe, 600 V (DC + AC peak); 1: 1 probe

Caxial Parts

	Model	Bandwidth	Impedance	Power rating
Termination	BP-50M1	DC to 3GHz	50 Ω	0.5W (Ave.), 500W (Peak)
	BP-75M1	DC to 3GHz	75 Ω	0.5W (Ave.), 500W (Peak)
	BB-50M1	DC to 1GHz	50 Ω	0.5W BNC(Ave.)
	BB-50M10	DC to 300MHz	50 Ω	5W BNC(Ave.)
Signal T	S-50S2	DC to 18GHz	50 Ω	1.25W SMA
Conversion pad	B-5P7J1	DC to 2GHz	50 Ω (Plug)/ 75 Ω (Jack)	0.5W BNC(Ave.)

FET Probes

Model	Attenuation	Input RC	Bandwidth
SFP-5A	10:1	Approx. 1.9pF, Approx. 1MΩ	DC to 1GHz
SFP-4A	10:1	Approx. 2.15pF, Approx. 1MΩ	DC to 800MHz
SS-220 (Attenuator) Custom Order	100:1	Approx. 1.9pF, Approx. 900kΩ	DC to 1GHz

Mini Clips

Mini clip HP-2

0.8 mm to 2.54 mm
Pipe length: 30 mm
10 colors



Test clip PB Series

0.08 mm to 0.23 mm
Pipe length: XX mm
(10 sets)



Probing arm T-60

Extremely useful when used with Mini clips.
Can be used with most probes.



Mini clip FP-7

0.3 mm to 1.25 mm
Pipe length: 1.3 mm
(10 sets)



Mini clip FP-7S

0.2 mm to 0.5 mm
Pipe length: 8 mm
(10 sets)



Mini clip FP-7L

0.3 mm to 1.25 mm
Pipe length: 14 mm
(10 sets)



Mini clip FP-7LV

0.3 mm to 1.25 mm
Pipe length: 14 mm
(10 sets)



Probe top conversion tool GR-DF

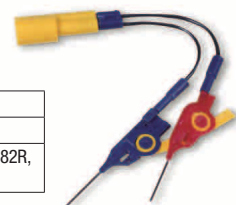
Comes with two FP-5L units
(or FP-7L units as an alternate choice)
Compatible with: SS-0110, SS-0112



Probe top conversion tool GR-AF/CF

Comes with two FP-5L units
(or FP-7L units as an alternate choice)

	Compatible
GR-AF	SS-0130R, 0131R
GR-CF	PP006/A/B, SS-101R, 103R, 082R, 0012, 0001, 0122



Optional probes for the isolation system

High Voltage Probes

Model	Bandwidth (MHz)	Attenuation ratio 1: x	Withstand voltage (duty 10%)	Withstand voltage (duty 50%)	DC	Length (m)	Remarks
PHV1000-R0	400	100	6kV (500ms cycle)	3.5kV (100ms cycle)	1kV	2	CAT I
PHVS1000-R0	400	1000	6kV (500ms cycle)	3.5kV (100ms cycle)	1kV	2	CAT I
PHV641-LR0	380	100	4kV (200ms cycle)	3.5kV (60ms cycle)	3kV (+AC peak)	1.2	AC rms 2kV
PHV661-LR0	380	100	6kV (200ms cycle)	5kV (60ms cycle)	4kV (+AC peak)	1.2	AC rms 2.8kV
PHV4002-3/0,6R0	100	1000	40kV (100ms cycle)	30kV (30ms cycle)	20kV (+AC peak)	3	AC rms 14kV



PHV1000-R0



PHV6xx-LR0



High Voltage Probe HV-P60

DC to 50MHz, DC to 60kV, pulse 80kV
The probe stand (SK-301) is optional.

Specifications and performance

Input/output ratio: 2000:1
Input RC: 1000MΩ,
5pF ± 2pF
DC to 50MHz



High Voltage Probe HV-P30

DC to 50MHz, DC to 30kV,
pulse 40kV

Specifications and performance

Input/output ratio: 1000:1
Input RC: 100MΩ,
5pF ± 2pF
DC to 50MHz



Current Probes SS-250

DC to 100MHz,
MAX 30A rms



Current Probes SS-240A

DC to 50MHz,
MAX 30A rms



Current Probes SS-270

DC to 2MHz,
MAX 500A rms



Current Probes SS-260

DC to 10MHz,
MAX 150A rms

Specifications and performance

Model	SS-250	SS-240A	SS-270	SS-260
Frequency Bandwidth	DC to 100MHz(-3dB)	DC to 50MHz(-3dB)	DC to 2MHz(-3dB)	DC to 10MHz(-3dB)
Maximum input range	30A rms	30A rms	500A rms	150A rms
Maximum input current	50A peak, non-continuous	50A peak, non-continuous 50A peak at pulse width ≤ 10 μs	700A peak, non-continuous	300A peak, non-continuous
Output sensitivity	0.1 V/A		0.01V/A	
Sensitivity accuracy	± 1.0% of reading ± 10mA for probes only (in the range from 0 to 30A rms, DC, and AC 45 to 66Hz)		± 1.0% of reading ± 5mV for probes only (in the range from 0 to 500A rms, DC, and AC 45 to 66Hz)	± 1.0% of reading ± 1mV for probes only (in the range from 0 to 150A rms, DC, and AC 45 to 66Hz)
	± 2.0% of reading for probes only (50A peak or less and over 30A rms, DC, and in the range of AC 45 to 66Hz)		± 2.0% of reading for probes only (700A peak or less and over 500A rms, DC, and in the range of AC 45 to 66Hz)	± 2.0% of reading for probes only (300A peak or less and over 150A rms, DC, and in the range of AC 45 to 66Hz)
Noise	2.5mA rms or less (observed with an 20MHz oscilloscope)		25mA rms or less (observed with an 20MHz oscilloscope)	
Power consumption	5.3VA max.	5.6VA max.	7.2VA max	5.5VA max
Measurable wire diameter	ø5mm		ø20mm	

POWER supply

PS-26

for SS-250/SS-240A/
SS-270/SS-260
(Input voltage
AC100V/120V/200V/240V)

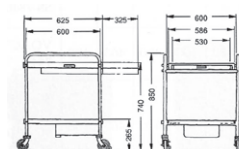


DS-579 [DS5500 series only]

for SS-250/SS-240A/
SS-270/SS-260



Scope Wagons MT-600



Digital Multimeter (portable type)

VOAC 7500H Series



Isolate 2-channel input dual function
0.1μV, 509999, 5½ digits

VOAC 7523H

Isolate 2-channel input dual function
1μV, 509999, 5½ digits

VOAC 7520H



4-terminal resistance measurement
dual function 0.1μV, 509999, 5½ digits

VOAC 7522H

4-terminal resistance measurement
dual function 1μV, 509999, 5½ digits

VOAC 7521H

Digital Multimeters VOAC7523H/7522H/7520H/7521H Specifications

* Accuracy (±X% of reading +Y digits) indicated by X+Y

The measuring accuracy indicated below can be obtained for a year following the calibration of the instrument.

1. Typical Sample Rate and Resolution

Sample Rate	Resolution	Reading Rate	Hum Rejection
SLOW	5.5-digit	approx. 4 times/sec	Yes
MID	5.5-digit	approx. 20 times/sec	Yes
FAST	4.5-digit	approx. 100 times/sec	N/A

2. DC Volt (DCV) 50mV range is for the VOAC7523H / 7522H only.

Range	Resolution		Input Resistance	Accuracy*	
	5.5-digit	4.5-digit		SLOW/MID	FAST
50mV	0.1μV	1μV	100MΩ or more	0.025+10	0.025+15
500mV	1μV	10μV	1000MΩ or more	0.012+5	0.012+10
5V	10μV	100μV	0.012+2	0.012+7	
50V	100μV	1mV		0.016+5	0.016+10
500V	1mV	10mV	approx. 10MΩ		
1000V	10mV	100mV		0.016+2	0.016+7

The accuracy in the 50mV and 500mV ranges is specified after zero compensation through the REL operation.

Sample rate in the 50mV range

SLOW/MID: Approx. 0.5 times/sec, FAST: Approx. 50 times/sec

Max. input voltage: 50mV to 5V range ± 800V (continuous)

50V to 1000V range ± 1100V (continuous)

Resolution and noise rejection

Resolution	Sample Rate	NMRR	CMRR
5.5-digit	SLOW	55dB or more	120dB or more
5.5-digit	MID	55dB or more	120dB or more
4.5-digit	FAST	0dB	55dB or more

3. CH-B DC Volt (DCV) VOAC7523H / 7520H only

Range	Resolution 4.5-digit	Input Resistance	Accuracy*	
			SLOW/MID	FAST
5V	100μV	CH-B:H to CH-B:L 10MΩ ± 3%	0.025+2	0.025+30
50V	1mV	CH-B:H to CH-A:L 5MΩ ± 3%		0.025+8
300V	10mV	CH-B:L to CH-A:L 5MΩ ± 3%		0.025+5

Max. input voltage: ± 300V, between CH-A L and CH-B ± 300V

Resolution and noise rejection

Resolution	Sample Rate	NMRR	CMRR	Isolation between CH-A and CH-B
4.5-digit	SLOW/MID	55dB or more	120dB or more	56dB or more
4.5-digit	FAST	0dB	55dB or more	

4. AC Volt (ACV, DC+ACV) detection of True RMS

Up to 100kHz for VOAC7521H / 7520H

Range	Resolution 5.5-digit	Measurement Range		Input Resistance
		SLOW	MID/FAST	
500mV	1μV	15Hz to 300kHz	200Hz to 300kHz	less than approx. 1MΩ // 100pF
5V	10μV			
50V	100μV			
500V	1mV	45Hz to 100kHz	200Hz to 100kHz	
750V	10mV	45Hz to 20kHz	200Hz to 20kHz	

Accuracy: SLOW Sample (Sine wave Amplitude at 5% to 100% of fullscale of range)

Frequency	Accuracy*
15Hz to 45Hz	0.5+150
45Hz to 100Hz	0.25+150
100Hz to 30kHz	0.2+150
30kHz to 100kHz	0.5+300
100kHz to 300kHz	2.5+1000

Coefficient to input other than sine wave

Crest Factor	Crest Factor		
	1 to 1.5	1.5 to 2	2 to 3
15Hz to 30kHz	0.05%	0.15%	0.30%
30kHz to 300kHz	0.20%	-	-

Response time

Sample Rate	Resolution	Reading Rate	Response Time
SLOW	5.5-digit	4 times/sec	less than 3 sec
MID/FAST	5.5-digit	20 times/sec	less than 2 sec

Max. input voltage: 780Vrms, ± 1100V DC (continuous)

In the case of DC+ACV, 500 (less than 45Hz) or 300 (45Hz or higher) must be added to the value of Accuracy digit in above.

Sample rate of FAST becomes the same values as MID (approx. 20 times/sec).

5. DC Current (DCA)

Range	Resolution		Accuracy*		Input Resistance
	5.5-digit	4.5-digit	SLOW/MID	FAST	
5mA	10nA	100nA	0.05+7	0.05+17	150Ω or less
50mA	100nA	1μA			15W or less
500mA	1μA	10μA			2W or less
10A	100μA	1mA	0.2+7	0.2+17	0.1Ω or less

Auto range is not available between 5mA to 500mA range and 10A range because of using different input terminals.

Max. input current: 500mA at 5mA to 500mA ranges (FUSE 0.5A/250V)

10A at 10A range (FUSE 15A/250V)

6. AC Current (ACA, DC+ACA)

Range	Resolution	Measurement Range		Input Resistance
	5.5-digit	SLOW/MID	FAST	
5mA	10nA	15Hz to 5kHz	200Hz to 5kHz	150Ω or less
50mA	100nA			15W or less
500mA	1μA	45Hz to 5kHz		2W or less
10A	100μA			0.1W or less

Accuracy: SLOW Sample (Sine wave) amplitude at 5% to 100% of fullscale (10% to 100% for 10A range)

Frequency	Accuracy*	Crest Factor		
		1 to 1.5	1.5 to 2	2 to 3
15Hz to 45Hz	1+200	0.05%	0.15%	0.30%
45Hz to 1kHz	0.4+200			
1kHz to 5kHz	5.0+200			

Response time

Sample Rate	Resolution	Reading Range	Response time
SLOW	5.5-digit	4 times/sec	less than 3 sec
MID/FAST	5.5-digit	20 times/sec	less than 2 sec

Max. input current:

500mA for 5mA to 500mA ranges (FUSE 0.5A)

10A for 10A range (FUSE 15A)

DC Component on input current must be included in the Max. input current.

In the case of 10A range at 45Hz to 1kHz, 0.3 must be added to %.

In the case of DC+ACA, 500 (less than 45Hz) or 300 (45Hz or higher) must be added to the value of Accuracy in above.

Sample rate of FAST becomes the same value as MID (approx. 20 times/sec).

7. Resistance (2 WireΩ / 4 WireΩ) 4 WireΩ: VOAC7522H / 7521H only

Range	Resolution		Accuracy*		Test Current
	SLOW/MID	FAST	SLOW/MID	FAST	
50Ω	0.1mΩ	1mΩ	0.025+10	0.025+15	approx. 10mA
500Ω	1mΩ	10mΩ			approx. 10mA
5kΩ	10mΩ	0.1Ω			approx. 1mA
50kΩ	0.1Ω	1Ω	0.014+3	0.014+8	approx. 100μA
500kΩ	1Ω	10Ω			approx. 10μA
5MΩ	10Ω	100Ω			approx. 1μA
50MΩ	100Ω	100Ω	0.25+30	0.25+30	approx. 100nA
500MΩ	1kΩ	1kΩ	1.5+50	1.5+50	approx. 10nA

Max. input voltage: ± 500V peak Open circuit test voltage: 12V or less

The accuracy at 50Ω to 5kΩ range are specified after zero compensation through the REL operation.

Sample rate of FAST at 5MΩ to 500MΩ range becomes the same value as MID (approx. 20 times/sec).

8. Low-Power Resistance (2 WireΩ)

Range	Resolution SLOW/MID/FAST	Accuracy*		Test Current
		SLOW/MID	FAST	
500Ω	10mΩ	0.1+5	0.1+15	approx. 1mA
5kΩ	0.1Ω			approx. 100μA
50kΩ	1Ω			approx. 10μA
500kΩ	10Ω	0.2+30	0.2+40	approx. 1μA
5MΩ	100Ω			approx. 100nA
50MΩ	1kΩ			approx. 10nA

Max. input voltage: ± 500V peak Open circuit test voltage: 12V or less

The accuracy at 500Ω to 5kΩ range are specified after zero compensation through the REL operation.

Sample rate of FAST at 5MΩ to 500MΩ range becomes the same value as MID (approx. 20 times/sec).

Indications are in 4.5 digits for SLOW, MID, and FAST.

9. Diode

Test Current	Measurement Range	Accuracy*	Open Circuit Test Voltage	Max. Input Voltage
approx. 1mA or 10mA	0.1mV to 5.0999V	0.014+13	12V or less	± 500V peak

10. Temperature

Thermocouple	Temperature Range to be Measured	Accuracy*	Resolution	Max. Input Voltage
R	-50°C to 0°C	0.2+70	0.1°C	± 500V peak
	0°C to +100°C	0.2+50		
	+100°C to +1768°C	0.2+30		
K(CA)	-200°C to -100°C	0.15+50		
	-100°C to 0°C	0.15+35		
	0°C to +1372°C	0.15+20		
T(CC)	-200°C to -100°C	0.15+50		
	-100°C to 0°C	0.15+35		
	0°C to +400°C	0.15+20		
J(IC)	-200°C to -100°C	0.15+50		
	-100°C to 0°C	0.15+35		
	0°C to +1200°C	0.15+20		
E(CRC)	-200°C to -100°C	0.15+50		
	-100°C to 0°C	0.15+35		
	0°C to +1000°C	0.15+20		

The above accuracy values do not include the thermocouple accuracy.

± 0.1°C/°C is added (for all thermocouples) with the operating ambient temperature of this product at 0°C to +18°C and +28°C to +50°C.

Accuracy is not specified. In the temperature range to be measured at -200°C or lower,

The reference thermo electric force is calculated by the "broken line approximation" according to JIS C 1602-1995.

If this product and the thermocouple are connected via a plug, an error for contact temperature compensation is added.

11. Frequency (AC couple, Crest Factor: less than 3)

Sample Rate	Reading Rate (Gate time)	Display Digits and Measurement Range	Accuracy*
SLOW	approx. 0.5 times/sec (1s)	6-digit 15.0000Hz to 1.00000MHz	0.02+2
MID	approx. 4 times/sec (100ms)	5-digit 15.0000Hz to 1.00000MHz	
FAST	approx. 10 times/sec (10ms)	4-digit 150.00Hz to 1.000MHz	

AC couple: Crest factor ≤ 3

AUTO range of ACV must be used with input attenuator.

Max. input voltage: 780 Vrms, ± 1100V peak

12. Chart for combination of Dual Function

	DCV	CH-B DCV ^(*)	ACV	DC+ACV	ACA	DC+ACA	2 WireW	4 WireW ^(*)	Hz	°C
DCV	△	△	△	△	△	△	△	△	△	△
CH-BDCV ^(*)	○	×	○	○	○	○	○	-	○	○
ACV	○	○	×	○	○	△	×	×	○	×
DC+ACV	△	○	○	×	○	△	×	×	○	×
ACA	△	○	○	○	×	△	△	△	△	×
DC+ACA	△	○	△	△	△	○	×	△	△	×
2 WireW	×	○	×	×	△	△	×	△	×	×
4 WireW ^(*)	×	-	×	×	△	△	△	×	×	×
Hz	△	○	○	○	○	△	×	×	×	×
°C	△	○	×	×	×	×	×	×	×	×

○: Available △: have a limitation ×: N/A -: not provided

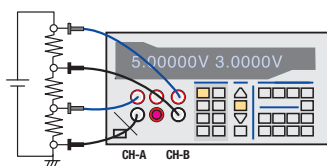
(*) CH-B DCV: VOAC7523H / 7520H only (**) 4 WireΩ: VOAC7522H / 7521H only

13. General

MATH	Moving Average, Scale, Decibel (dBm, dBu), Difference, Statistics (MAX, MIN, X, s), Comparison (COMP), Arithmetic Calculation between Dual Function
Memory	Max. 3000 data with 10 msec resolution time mark (Elapsed time)
Interfaces	Standard RS-232
(Full Remote)	Option LAN, GPIB
Power Supply	Voltage AC100V, 110V, 220V, 240V
Frequency	50Hz, 60Hz
Power Consumption	21 VA (includes options) or less

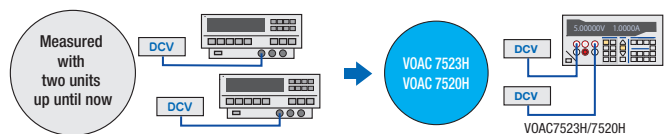
Isolate 2-channel input (VOAC 7523H/7520H)

- If the CH-A and CH-B input is from an insulated VOAC 7523H or 7520H, the electrical potential for different circuits can be measured simultaneously.

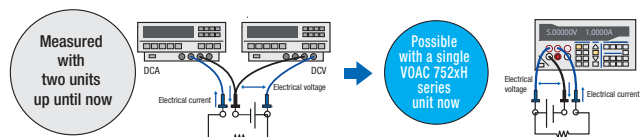


Dual Channels

- Measurements that conventionally require two oscilloscopes can now be performed simultaneously with a single unit to greatly improve efficiency. A connection example is shown below. Simultaneous display and simultaneous measurements are being performed here



Dual Display / Dual Function



Accurate Root-Mean-Square (RMS)

- Accurate root-mean-square values for AC voltage and AC current can be measured. Root-mean-square values for direct current can also be measured (DC+AC) V, (DC+AC) A

Options

Product Name	Part Number	Image of Product
LAN interface *1	SC-351	
DIO interface *2	SC-352	
GPIB interface *1	SC-353	
D/A Converter interface *2	SC-354	
RS232-USB Converter (WindowsXP, Vista, 7)	SC-525	
4-wire kelvin test leads	SC-005	
High-voltage probe 30kVdc max. (can be used only in the Zin of 10MW range)	SC-003	
Clamp-on current probe DC ± 200A MAX AC150A rms MAX (40Hz to 500Hz)	SC-011	
Sheath-type thermocouple (Type K) -200°C to +800°C	SC-0107	
Surface thermocouple (Type K) 0°C to +500°C	SC-0116	

Operation Temperature and Humidity	0°C to +50°C (less than 80%RH) no condensation, 70%RH or less at +40°C to +50°C
Storage Temperature and Humidity	-20°C to +60°C (70%RH or less) no condensation, includes operation temperature
Size	Measurement 210W x 99H x 353L mm (Options are built into the main unit)
Weight	3.5kg (includes options) or less
Standard Accessories	Fuse, Test Leads, Alignment Screwdriver, Operation Manual(CD-ROM), Power cable

Abundant Interfaces

• LAN Interface: SC-351

10BASE-T (cannot be connected at the same time as the GPIB)

• GPIB Interface: SC-353

To create a familiar system

• DIO Interface: SC-352

Useful for judging acceptable and non-acceptable waveforms. Open collector output.

• D/A Output: SC-354

Output can be selected from three patterns of 10V, 1V and 0.1V.

Cannot be connected at the same time as the DIO.

See the following website for further details.

www.iti.iwatsu.co.jp/products/voac/voac752xh_opt.html

• RS USB Converter: SC-525

USB can be used when connected with a RS-232 connector.

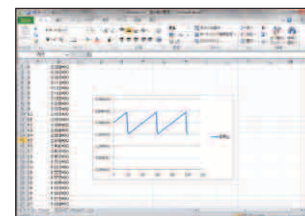
Trend Graphs Using the Interface

Data can be loaded into Excel and other spreadsheet software when connected to a PC with the interface. This enables trend graphs, etc., to be easily made.

Programming not required!
Download the software from the following.

Click here for the sample software

URL <http://www.iti.iwatsu.co.jp/> → Support



Product Name	Part Number	Image of Product
Banana plug (Can be used to connect a thermocouple)	POMONA1286	
High-resistance test lead	SC-004	
Test leads	SC-020	
Arrow clip For SC-020 (AC30V/DC60V/DC3A)	SC-021	
Alligator clip For SC-020 (AC30V/DC60V/DC10A)	SC-022	
Alligator clip H For SC-020 (600Vrms, CAT II/10A)	SC-023	

*1 The LAN interface SC-351 and GPIB interface SC-353 cannot be installed at the same time.

*2 The DIO interface SC-352 and D/A Converter interface SC-354 cannot be installed at the same time.

Digital Multimeters (Handy type)

1μV, 50000, 4½ digits
VOAC 22



100μV, 9999, 4-digit at full scale
VOAC 87



Universal Counter SC-7200H Series

A new lineup of high-performance counters that transcend their class!



GPIB **CE**

3GHz x 1ch &
230MHz x 2ch
Universal Counter
SC-7207H



GPIB **CE**

2GHz x 1ch &
230MHz x 1ch
Universal Counter
SC-7206H



GPIB **CE**
[Option]

230MHz x 2ch
Universal Counter
SC-7205H

Useful functions based on the need for a maximum of 3GHz and easy use.

- Enables frequency measurements for two independent channels (SC-7207H, SC-7205H.)
- Pulse width measurements and time interval measurements greatly broaden the scope of single-gate measurement.
- Easy operations with single key strokes for each action.
- Easy-to-see fluorescent display area. Detailed information displayed with 5 x 7 dot resolution.
- * A full-spelling guide provides powerful support for operations.
- Auto-trigger function that eradicates the need for setting the trigger level. Manual setup is, of course, also possible.
- Making line inspection tasks more efficient is a simple chore with the comparison and statistic calculation functions.
- The scaling calculation function enables single unit conversion (revolutions, speed, etc.)
- Input signal peak voltage measurements make it easy to confirm the waveform amplitude.
- The save/recall function for panel setup makes predetermined inspection tasks more efficient.
- The GPIB (optional for the SC-7205H: SC-701) and RS-232 interfaces provide full remote control.

* Transmission is performed in the real-time at a high speed of a maximum 200 items of data/second, which contributes to improved line throughput.

- Full lineup of options to provide greater expandability
- Comparator output (open collector) with digital I/O (SC-702.) External trigger input.
- * 150mA can be used for line monitoring equipment without modification to provide a margin of 50V.
- The high-stability standard oscilloscope (SC-703A) provides highly accurate measurements.

Specifications and Performance

Output Interfaces	RS-232: Fitted as standard. GPIB Fitted as standard (optional for the SC-7205H: SC-701) Digital I/O: Optional (SC-702)
Size and Weight	Approximately 210W x 99H x 353L mm (excluding options and protrusions) 4.0kg or less (when mounted with the SC-701, 702 and 703 options)
High-stability Standard Oscillator (manufactured on request)	Two types of options available (only one type may be mounted) Temperature Characteristics: +/-0.05ppm, Oscillation Frequency: 10MHz

Universal Counter Option

GPIB Interface

SC-701

 For use with the SC-7205H

- Mounting the SC-701 onto the SC-7207H, 7206H and 7205H Universal Counters (fitted as standard to the SC-7207H and 7206H) enables measurements taken with external GPIB controllers to be reset, the remote setup of measurement functions, time base functions and calculations, etc., and the results of measurements to be transmitted as data to external sources.

* This is a factory option and needs to be ordered at the same time as the main unit.
Ordering factory installation at a later date will be chargeable.

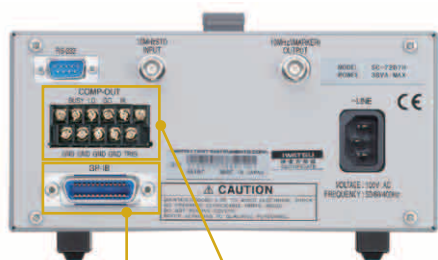
Digital I/O

SC-702

 For use with the SC-7207H, SC-7206H and SC-7205H

- Installing the SC-702 onto the SC-7207H, SC-7206H and SC-7205H Universal Counters will enable control over the start of measurement and the output of comparison calculation results. (open collector) Connecting an external lamp also allows parts to be selected and inspection results to be easily browsed.

* This is a factory option and needs to be ordered at the same time as the main unit.
Ordering factory installation at a later date will be chargeable.



SC-701

SC-702

Digital I/O (Factory option)

GPIB Interface (Equipped as standard on SC-7207H and SC-7205H. Factory option only for SC-7206H)

Main Performance

Maximum Output Terminal Rating	Withstand voltage	DC50V
	Withstand current	DC150mA
	Frequency response	DC to 1kHz
Maximum Input Terminal Rating	Withstand voltage	DC5V
	Frequency response	DC to 1kHz

RS-USB Converter

SC-525

For use with the SC-7207H, SC-7206H and SC-7205H

- The cable for connecting the RS-232 measurement unit to a personal computer's USB port.
- Overall length approximately 85cm.

* Can also be used with the VOAC 7500H series, the SG-4115 and the SG-4105.



High-stability Standard Oscilloscope

SC-703A

 Custom Order

	New Crystal (SC-703A)
Oscillation Frequency	10MHz
Temperature Characteristics	+/-0.05ppm Range of 0 °C to 40 °C with +25 °C as the standard.
Rising Time	+/-0.05ppm 10 minutes for power switch-on with the frequency 1 hour after power switch-on as the standard
Time Fluctuations (per day)	+/-0.02ppm Value at 72 hours after power switch-on with 48 hours after power switch-on as the standard
Time Fluctuations (per year)	+/-0.02ppm Value at one year after power switch-on with 10 days after power switch-on as the standard

Universal Counters SC-7207H / SC-7206H / SC-7205H Specifications

Frequency A (FREQ-A)					
●Measuring range and resolution * SC-7206H is not equipped with EXT-B gate					
		SC-7207H		SC-7206H, SC-7205H	
Reference time (reference frequency)		10ns (100MHz)		100ns (10MHz)	
Range		6mHz to 230MHz		0.6mHz to 230MHz	
		10Hz to 230MHz			
Resolution and count method	DC	Below 100MHz	100MHz or more	Below 10MHz	10MHz or more
	AC	Reciprocal count	Direct count	Reciprocal count	Direct count
	Frequency	5 digits	1kHz	4 digits	1kHz
	Count method	6 digits	100Hz	5 digits	100Hz
	1ms gate	7 digits	10Hz	6 digits	10Hz
	10ms gate	8 digits	1Hz	7 digits	1Hz
	0.1s gate	9 digits	0.1Hz	8 digits	0.1Hz
	1s gate	Reciprocal count method: The number of digits is determined by external gate time			
	10s gate	Reciprocal count method: The number of digits is determined by measured signal			
	EXT-B gate *				
SGL gate					

AC Line Frequency (FREQ-LINE) (for SC-7207H and SC-7205H only)			
Measuring range and resolution			
Reference time		SC-7207H	SC-7205H
		10ns	100ns
Range		45Hz to 440Hz	
Resolution	0.1s gate	7 digits	6 digits
	1s gate	8 digits	7 digits
	10s gate	9 digits	8 digits

Frequency C (FREQ-C) (for SC-7207H and SC-7206H only)					
Measuring range and resolution					
Reference time (reference frequency)		SC-7207H		SC-7206H	
		10ns (100MHz)		100ns (10MHz)	
Range (for AC coupling only)		100MHz to 3GHz 1/16 prescaler		100MHz to 2GHz 1/16 prescaler	
Resolution and count method	Measured signal	Below 1.6GHz	1.6GHz or more	Below 160MHz	160MHz or more
	Count method	Reciprocal count	Direct count	Reciprocal count	Direct count
	1ms gate	5 digits	10kHz	4 digits	10kHz
	10ms gate	6 digits	1kHz	5 digits	1kHz
	0.1s gate	7 digits	100Hz	6 digits	100Hz
	1s gate	8 digits	10Hz	7 digits	10Hz
	10s gate	9 digits	1Hz	8 digits	1Hz
	EXT-B gate	Reciprocal count method: The number of digits is determined by external gate time		Not equipped with EXT-B	

Period A (PERI-A)			
Measuring range and resolution *SC-7206H is not equipped with EXT-B gate			
Reference time		SC-7207H	SC-7206H, SC-7205H
		10ns	100ns
Range		5ns to 171s	5ns to 1,717s
Resolution	DC couple	5ns to 0.1s	
	AC couple		
	1ms gate	5 digits	4 digits
	10ms gate	6 digits	5 digits
	0.1s gate	7 digits	6 digits
	1s gate	8 digits	7 digits
	10s gate	9 digits	8 digits
	EXT-B gate *	The number of digits is determined by external gate time	
	SGL gate	The number of digits is determined by measured signal	

Duty ratio A (DUTY-A)			
Measuring range and resolution			
Input signal frequency range		SC-7207H	SC-7206H, SC-7205H
		Same as FREQ-A	
Measuring range	SGL gate	0.01μ to 99,999,999,99 [%]	
	Internal gate	0.2μ to 99,999,999,8 [%]	2μ to 99,999,998 [%]
	SGL gate	10ns/input period x 100 [%]	100ns/input period x 100 [%]
	1 to 24	10ns/average input period x 100 [%]	100ns/average input period x 100 [%]
	25 to 2,499	1ns/average input period x 100 [%]	10ns/average input period x 100 [%]
	2,500 to 249,999	100ps/average input period x 100 [%]	1ns/average input period x 100 [%]
	250,000 to 24,999,999	10ps/average input period x 100 [%]	100ps/average input period x 100 [%]
	25,000,000 or more	1ps/average input period x 100 [%]	10ps/average input period x 100 [%]

Pulse width A (P.W-A)			
Minimum pulse width: 6ns Maximum repetitive frequency: 80MHz Measuring range and resolution			
Reference time		SC-7207H	SC-7206H, SC-7205H
		10ns	100ns
Measuring range		10ns to 171s	100ns to 1,717s
Measuring resolution	Internal gate (1ms to 10s)	10ns to approx. 1/2 gate time	100ns to approx. 1/2 gate time
	SGL gate	10ns to 100ns	100ns to 1ms
	1 to 24	10ns	100ns
	25 to 2,499	1ns	10ns
	2,500 to 249,999	100ps	1ns
	250,000 to 24,999,999	10ps	100ps
	25,000,000 or more	1ps	10ps

Time interval A --> B (T.INT A --> B) (for SC-7207H and SC-7205H only)			
Minimum time interval: 6ns Maximum repetitive frequency: 80MHz Measuring range and resolution			
Reference time		SC-7207H	SC-7205H
		10ns	100ns
Measuring range		10ns to 10,955s	100ns to 109,951s
Measuring resolution	Internal gate (1ms to 10s)	10ns to approx. 1/2 gate time	100ns to approx. 1/2 gate time
	SGL gate	10ns to 10μs	100ns to 100μs
	1 to 24	10ns	100ns
	25 to 2,499	1ns	10ns
	2,500 to 249,999	100ps	1ns
	250,000 to 24,999,999	10ps	100ps
	25,000,000 or more	1ps	10ps

Frequency ratio A/B (FREQ A/B) (for SC-7207H and SC-7205H only)				
•Measuring range and resolution				
			SC-7207H	SC-7205H
Input signal frequency range			Both CH-A and CH-B are the same as that for FREQ-A	
Measuring range		Internal gate (1ms to 10s)	1E-9 to 1E+9	
Measuring resolution		Internal gate (1ms to 10s)	1+LOG (CH-A input frequency x gate time) digits	
Phase measuring A --> B (PHAS A --> B) (for SC-7207H and SC-7205H only)				
•Minimum time interval: 6ns •Maximum repetitive frequency: 80MHz •Measuring range and resolution				
			SC-7207H	SC-7205H
Reference time			10ns	100ns
Measuring range		SGL gate	0.1μ to 359,999,999 □	
		Internal gate	1μ to 359,999,999 □	
Measuring resolution	Average count of internal gate	SGL gate	0ns/input period x 360 □	
		1 to 24	10ns/average input period x 360 □	
		25 to 2,499	1ns/average input period x 360 □	
		2,500 to 249,999	100ps/average input period x 360 □	
		250,000 to 24,999,999	10ps/average input period x 360 □	
		25,000,000 or more	1ps/average input period x 360 □	
Peak voltage measuring (SC-7206H is not equipped with CH-B)				
Measures and displays in real-time the voltage amplitude of the measured signal at CH-A or CH-B.				
Frequency range			150Hz ≤ input frequency ≤ 50MHz	
Response time			2 seconds or less	
Voltage range			± 2.50V (ATT off, resolution: 10mV), ± 50.0V (ATT on, resolution: 100mV)	
Measuring error			ATT off: 10% of indication ±50mV ATT on: not specified	
CH-A, CH-B input terminal (SC-7206H is not equipped with CH-B)				
Input RC			Approx. 1MΩ//20pF or less	
Coupling			AC or DC	
Low pass filter			Off, 10kHz	
Attenuator			Off, 26dB (1/20)	
Trigger level	Measuring Range	ATT off	-2.50V to +2.50V (resolution: 10mV)	
		ATT on	-50.0V to +50.0V (resolution: 100mV)	
	accuracy (0°C to +40°C)	ATT off	10% ± 30mV of the set value (± 3% when +2V to -2 V)	
		ATT on	10% ± 300mV of the set value (± 3% when +40V to -40 V)	
Operating input voltage range			± 2.5V ± 50V	
Input sensitivity	Manual trigger	ATT off	30mVrms (DC to 230MHz)	
		ATT on	0.6Vrms (DC to 230MHz)	
	Auto trigger	ATT off	200mVrms (10kHz to 230MHz, sine wave)	
		ATT on	4Vrms (10kHz to 230MHz, sine wave)	
CH-C input terminal (for SC-7207H and SC-7206H only)				
Maximum input power			+30dBm (approx. 7Vrms when 1mΩ/50Ω = 0dBm as a reference)	
Impedance			Approx. 50Ω	
Coupling			AC	
VSWR			2.0 or less (SC-7207H: 100MHz to 3GHz, SC-7206H: 100MHz to 2GHz)	
Input sensitivity			(Sine wave: up to 2 GHz for SC-7206H) (100MHz ≤ input frequency ≤ 300MHz)	
AGC off/on		-20dBm	(300MHz < input frequency ≤ 1.5GHz)	
		-25dBm	(1.5GHz < input frequency ≤ 3.0GHz)	
		-20dBm		
Burst detection	Detection frequency range		SC-7207H	SC-7206H
			100MHz to 3GHz	100MHz to 2GHz
	Input sensitivity		(Sine wave: up to 2GHz for SC-7206H) (100MHz ≤ input frequency ≤ 1.2GHz)	
	AGC off	-20dBm	(1.2GHz < input frequency ≤ 3.0GHz)	
		-10dBm		
Detection delay time		500μs (Burst period ≥ set gate + 500μs)		
10MHz STD IN				
BNC terminal for more stable input of the external reference frequency				
Frequency			10MHz± 50Hz (± 5ppm)	
Amplitude			1Vrms to 5Vrms, threshold = 0V	
Input resistance			Approx. 6.4kΩ	
Input coupling			AC	
10MHz STD OUT/(MARKER OUT)				
BNC terminal for output of internal reference oscillator or marker signal.				
Marker signal is a signal that presupposes the brightness modulation (Z axis) of the analog oscilloscope for example. It is enabled at the SGL gate when the function is in between the time interval (T.INT A --> B) and phase (PHAS A --> B). Output is "Lo level" from the start of CH-A measuring to the start of CH-B measuring.				
Output		CMOS level		
Reference frequency output		10MHz: Stability is the same as that for the internal reference oscillator.		
Marker output		In the 5MHz band, L-state is output during actual measuring. (for SC-7207H and SC-7205H only.)		
Output interface			Environmental conditions	
•RS-232 is equipped as standard •GPIB is equipped as standard (option SC-701 for SC-7205H) •Digital I/O option can be installed (SC-702)			•Warm-up time: 60 minutes or more •Operating temperature/humidity: 0°C to +40°C/85%R.H or less (no condensation) •Storage temperature/humidity: -20°C to +60°C/90%R.H or less (no condensation)	
Reference oscillator				
Equipped with SC-7207H, SC-7206H and SC-7205H as standard				
Output is possible to the 10MHz OUT BNC terminal on the rear panel of the main unit.				
•Oscillation frequency: 10MHz •Temperature characteristics: ± 2.5ppm/unit environmental temperature: 0°C to +40°C •Aging rate: ± 1.0ppm/year				
Power supply conditions and power supply voltage changes (factory option)				
•Voltage: AC100V / 110V to 120V / 220V to 240V •Frequency: 50Hz, 60Hz, 400Hz •Power consumption: At AC100V with optional SC-701 and SC-702 are installed.				
		SC-7207H	SC-7206H	SC-7205H
Power Consumption		36VA MAX	33VA MAX	31VA MAX
Size		(210 ± 2)W x (99 ± 2)H x (353 ± 2)L mm (excluding options and protruded parts)		
Weight		4.0kg or less (including optional SC-701 and SC-702)		
Accessory		Power cable (1), operation manual CD-ROM (1)		

Universal Counter
SC-7217



A Maximum of 3GHz, and the Digit Display Greatly Increased to Accommodate a Maximum of 12Digits/sec

- USB, LAN, RS-232 (option) and full remote control with GPIB (option)
- Compare output with digital I/O
- Full lineup of options to provide greater expandability
 - Data stored on USB storage memories.
 - High-stability clock oscillator option.

SC-7217 Interim Specifications

CH-A, CH-B	Input impedance		50 Ω ± 1.5% / 1MΩ ± 1.5% //16pF ± 3pF		
	Input withstand pressure		50 Ω / 1MΩ		
	Frequency band		DC / AC		
	Input voltage range		DC to 450MHz / 10Hz to 450MHz		
	Trigger level accuracy		± 2.5V / ± 50V		
	Slope switching		± 2% ± 25mV / ± 2.5% ± 500mV		
	Band limiter		+ / -		
	Noise rejection		10kHz		
EXT-B	Input signal range		Pulse width / frequency	500ns min / 1MHz max	
CH-C	Input impedance / SWR / Maximum input electrical power			50 Ω, AC coupling / 2.0 or less / +30dBm	
	Frequency band			100MHz to 3GHz	
	AGC			ON/OFF	
	Burst detection			ON/OFF	
	Detection sensitivity			Up until 1.2GHz: -20dBm, up until 3GHz: -10dBm	
	Burst detection delay time			10μs	
Measurement Functions	FREQ A, FREQ B		Max. 13-digit, 12-digit/sec (at 1second gate)		
	Measurement range		Single: 6mHz to 250MHz, time / EXT-B gate: 12mHz to 450MHz		
	Gate selection		Single / EXT-B / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	FREQ C		Max. 13-digit, 12-digit/sec (at 1second gate)		
	Measurement range		100MHz to 3GHz, 1/16 pre-scaler		
	Gate selection		EXT-B / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	FREQ LINE		Measurement range / Gate selection		
	PERIOD A		45Hz to 440Hz / 0.1s/1s/10s		
	Measurement range		Single: 4ns to 166s		
	Gate selection		Time / EXT-B gate: 2.2ns to 83s		
	DUTY A		Single / EXT-B / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	Input signal range		Pulse width / Frequency	6ns min / 80MHz max	
	Measurement range		Single / Time	0.01μ to 99.999,999,99% / 0.2μ to 99.999,999,8%	
	Gate selection		Single / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	PULSE WIDTH A		Input signal range		
	Measurement range		Single / Time	6ns min / 80MHz max	
	Gate selection		6ns to 171s / 6ns to approximately ½ gate time		
	TIME INTERVAL A→B		Single / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	Input signal range		Pulse width / Frequency	6ns min / 80MHz max	
	Measurement range		Single / Time	6ns to 10,995s / 6ns to approximately ½ gate time	
	Gate selection		Single / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	FREQ A/B		Input signal range, Frequency		
	Measurement range / Gate Selection		250MHz max		
	PHAS A→B		1 E-9 to 1 E+9 / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	Input signal range		Pulse width / Frequency	6ns min / 80MHz max	
	Measurement range		Single / Time	0.1μ to 359.999,999,97 / 1μ to 359.999,999" (However, it is necessary or this to be less than half of the gate for non-measurable signal cycles)	
	Gate selection		Single / Time (set at 10μs to 10s 10 ⁿ) (n: integer)		
	TOT A		Input signal range		
	Gate selection		Pulse width / Frequency	2ns min / 250MHz max	
Measurement range		MANUAL / EXT-B / Time (set at 10μs to 10s 10 ⁿ) (n: integer)			
Peak voltage measurement		0 to 4,294,967,295 count			
Measurement frequency / Measurement speed		150Hz to 150MHz / 2 seconds or less			
Measurement voltage range		ATT OFF / ATT ON			
Measurement operations				± 2.5V / ± 50V	
Calculation				Repeat / Single / HOLD	
Pulse setup				Smoothing (moving average), scaling, comparate, statistics (MAX, MIN, σ , average)	
DATA save memory				Internal memory (10) or USB memory	
Internal standard clock				MAX. 500,000kinds (volatile memory)	
Temperature characteristics				+/- 1ppm (range of 0 to +40°C with +25°C as the standard)	
Temporal change / Short-term stability				± 0.1 ppm/month、 ± 1 ppm/year / ± 1 ppb/s	
Interface				USB / LAN / DIO	
USB / LAN / DIO				USB2.0 HS / 100base-TX / Output: HI/LO/GO/BUSY	
10MHz STD IN				Approximately 850 Ωs (at 10MHz), AC coupling / 10MHz +/- 50Hz / 100mVrms	
Marker / STD output				STD / Marker selected and output with the setting	
Output impedance / Marker output / STD output				50 Ωs +/- 10% / +1Vo-p (0V output during measurement) / 10 MHz sine wave 1Vp-p or more (with 50 Ωs at the terminal)	
Options (OP when shipped)	OCXO	Medium stability	Temperature characteristics	+/- 20ppb (range of 0 to +40°C with +25°C as the standard)	
			Temporal change	+/- 10ppb/day (fluctuations in one day` s frequencies with the standard frequency being that measured after 48 hours. At +25°C)	
		High stability	Temperature characteristics	+/- 100ppb/year (fluctuations in one year` s frequencies with the standard frequency being that measured 10 days after the power has been switched on. At +25°C)	
			Temporal change	+/- 5ppb (range of 0 to +40°C with +25°C as the standard)	
				+/- 0.5ppb/day (fluctuations in one day` s frequencies with the standard frequency being that measured 30 days after the power has been switched on. At +25°C)	
				+/- 50ppb/year (fluctuations in one year` s frequencies with the standard frequency being that measured 30 days after the power has been switched on. At +25°C)	
	Interface				GPIB (conforming to IEEE488-1 with full remote functions,) RS-232C, host for connecting the USB memory (for storage only)
	Electric power	Voltage / Frequency			AC 100V to AC 240V ± 10% / 50 to 60Hz ± 5%(100V to 240V) / 400Hz ± 10%(100V to 120V)
Power consumption			70VA(35W) max		
External dimensions (W x H x D)				(210 ± 2) × (99 ± 2) × (353 ± 2) mm	
Accessories				Product users` guide x 1, instructions (CD) x 1, power cable x 1.	
Environment				0°C to +40°C with 80%RH or less and no condensation	

Function Generator

SG-4100 Series



10mHz - 15MHz 1ch
SG-4105



10mHz - 5MHz 1ch
SG-4104

- Wide oscillation bandwidth from 10mHz to 15MHz (SG-4105)
- High accuracy (50ppm: SG-4105, SG-4104) and high definition waveform output by employing DDS (Direct Digital Synthesizer) system
- Max 20Vp-p (Output terminal open) (SG-4105)
- 0.0% ~ 100.0% duty control/ Up to 65,536 Burst waveforms
- Offset control +10V ~ -10V (output terminal open)
- Waveform outputs are connected continuously when vary the frequency
- Linear / Log sweep function
- Simultaneous display of the frequency and output voltage
- Easy operation (set performance can be checked at a glance)
- PMC option (SG-506: SG-4105) best suited for evaluating pulse motor control
- Provides Small-amplitude on Large-offset

PMC function*(Factory option)

Pulse motor control function SG-506 (SG-4105)

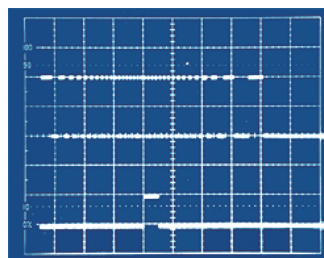
Pulse motor control function

PMC option function controls pulse motor. Pulse motor acceleration or braking controls need to be reviewed not only by position control, but also under loaded condition. The PMC option simplifies the evaluation. Pulse outputs in open collector (50V) are output from rear panel with PMC option.

Common motor driver circuit connected with PMC.

*PMC (Pulse Motor Control) is coined word by IWATSU TEST INSTRUMENT CORPORATION.

(Order any factory options when ordering the main unit. Additional orders after the delivery of the main unit require a separate fee.)



Upper waveform shows drive pulse for pulse motor, lower waveform shows sensor input waveform. After reaching maximum frequency while specified accelerating period, starts braking by sensor input signal. Then stops at specified pulse counts.

Boost Amp

SG-300

A useful drive amp that boosts signal generator output at 1MHz full power band.

The SG-300 is an amplifier for converting function generator output.

This amplifire can be used for a wide range of purposes, including the development of inverters and other mechatronic equipment.

The amplifire has a low impedance (Lo Ω) output, which enables it to be used with low power loss even driving low impedance loads.

It also supports amplitude modulation only at the positive side or only at the negative side, which enables zero level adjustment.

SG-300 Specifications

Maximum Voltage	24Vp-p (with 50 Ω loads) / 48Vp-p (without any load)
Maximum Current	DC or Peak 240mA (with 50 Ω loads) / Continuous DC or Peak 300mA (with Lo Ω output)
Full Power Band	1MHz (with a 50 Ω load and 24Vp-p output)



RS-USB Converter

SC-525

For use with the SG-4105

- The cable for connecting the RS-232 measurement unit to a personal computer's USB port.

- Overall length approximately 85cm.

* Can also be used with the VOAC 7500H series, SC-7200H series.



SG-4100 Series Specifications

Model			SG-4105		SG-4104	
Output waveform			Sine wave, square wave, pulse wave, triangle wave, ramp wave, DC output			
Output channels			1			
Output impedance			50Ω fixed			
Frequency	Sine, square		10mHz-15MHz		10mHz-5MHz	
	Pulse		10mHz-100kHz(without DC)			
	Others		10mHz-100kHz (without DC)			
	Accuracy		± 50ppm			
	Resolution		10mHz or 5 digits			
Amplitude	Range		50mVp-p to 10Vp-p (50 Ω) 100mVp-p to 20Vp-p (open)			
	Accuracy (sine 1kHz)		± 1%			
	Resolution		Except for DC waveform output : 0.1mV or 3 digits(50.0mV to 10.0V) For DC waveform output : 1mV or 3digits(1mV to 5.00V)			
	Flatness(reference to 1kHz)		Sine: up to 100kHz ± 1%		± 2%	
			Over 100kHz-1MHz ± 2%		± 3%	
			Over 1MHz ± 3%		± 5%	
			Square: up to 100kHz ± 3%, over 100kHz ± 5%			
		Triangle, pulse: ± 3% SLOW mode (all waveforms)				
Mode		CONT, TRIG, GATE				
Offset	Range		± 5V (waveform + DC offset less than 5.025V)			
	Resolution		1mV or 3digits			
	Accuracy		± 1% setting value ± 5mV (at DC output)			
Duty	Square wave		40-60% (more than 5MHz) 20-80% (5MHz or less)		20-80%	
	Others		0-100% (pulse wave, ramp wave)			
Sine wave spectral purity *Amplitude at 10Vp-p	Harmonic distortion *		Up to 100kHz -50dBc Over 100kHz-1MHz -45dBc Over 1MHz-15MHz -35dBc (SG-4104: up to 5MHz)			
	Spurious *		100kHz-1MHz -60dBc 1MHz-10MHz -50dBc 10MHz-15MHz -40dBc (SG-4104: up to 5MHz)			
	Total harmonic distortion		0.10% at 1Vp-p or more, less than 100kHz 0.30% at less than 1Vp-p, 100kHz			
Step response	Overshoot		Square 2% (less than 1MHz) Pulse 2% (1μs or more FWHM *) *Full Width at Half Maximum			
Rise, Fall time (10%-90%)	Square		20ns or less		35ns or less	
	Pulse		300ns or less			
SWEEP	Curve		Linear/Log		—	
	TYPE		CONTINUOUS: Oscillation continues at STOP frequency STOP: Oscillation stops after STOP frequency			
	Mode		CONT, GATE, TRIG			
	SWEEP TIME		1ms-500s			
	Sync out		SWEEP SYNC, SWEEP MARKER			
BURST	Number		65536		—	
	Mode		CONT, TRIG selectable ON counts or OFF counts			
PHASE	Range		± 359.9°			
	Resolution		0.1°			
Pulse motor modulation (available only when PMC option is installed) BG-506	Oscillation frequency		10mHz-100kHz		—	
	Waveform		Pulse			
	Control		CW(one-way control)			
	Output		Open collector 1 output			
	Sensor input		IN1, IN2			
	Mode		LIN-A,: After acceleration, braking by sensor input LIN-B: After acceleration, braking by specified pulse counts			
	Acceleration, braking curve		Linear			
	Pulse counts		16777216			
	Acceleration, braking time		1ms-45s			
TRIG	Level, polarity		TTL level (H: 2.1V min., L: 0.9V max.)			
	Inputs		1			
	Input impedance		1kΩ or more			
	Minimum pulse width		100ns			
	TRIG delay	Normal mode	Square: 5% of period +350ns or less Pulse: Period/16384 + 800ns or less			
Setup memory			9			
Remote control			GPIO, RS-232		—	
Power	Voltage		AC100V± 10% AC110V, 220V, 240V (factory option)			
	Frequency		50Hz, 60Hz, 400Hz			
	Consumption		Approx. 55VA			
Dimensions			Approx. 210W x 99H x 353L mm			
Weight			Approx. 4kg			
Environment conditions	Storage		-20°C to +60°C/90%R.H. or less			
	Operating		0°C to +40°C/85%R.H. or less			
Warm up time			30 min. or more			
Accessories			Power cable (1), operation manual (1)			
Factory option			PMC: SG-506		—	

* Waveform selection and frequency for CH1 and CH2 are the same.
SWEEP, modulation and BANK functions are disabled.

Delay Pattern Generator (6 channel pulse generator) DG-8000



Seamless change

The frequency, pulse width, and other settings can be seamlessly changed during oscillation.

Tracking function

Parameters can be changed simultaneously for each channel.

Operation pattern control (DG-802)

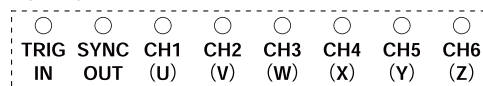
The operation pattern option enables continuous operation testing.

Synchronization of multiple generators (DG-602)

The quick synchronization option enables three generators (18 channels) to synchronously output data.

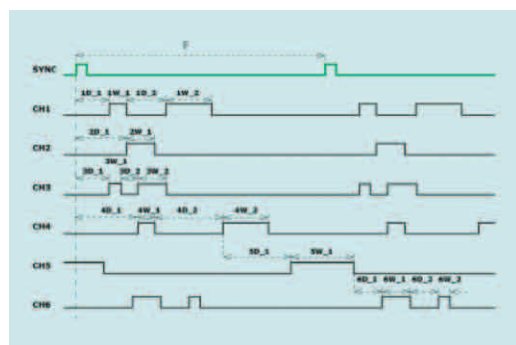


*input/output on the front



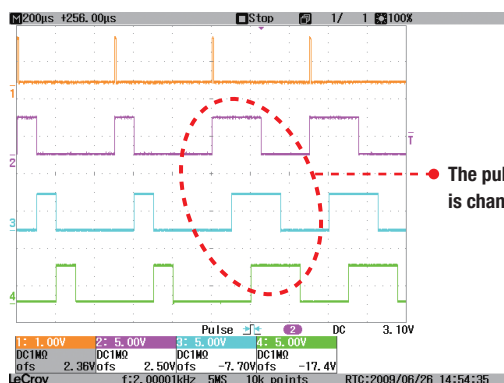
Rear panel configuration of a standard model

Setting parameters and output examples of 6 channel independent pulse output



BASIC mode

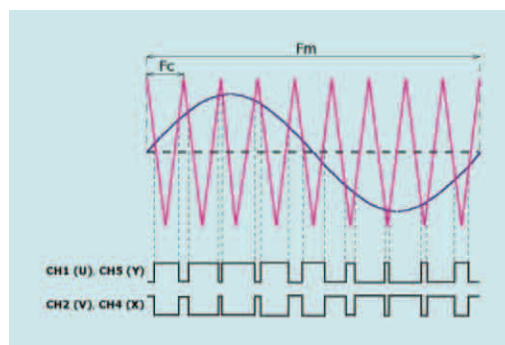
Pulses can be easily generated by specifying any dependency, delay value, and width value for each of 6CH. The output level can also be individually specified for each CH.



Tracking function

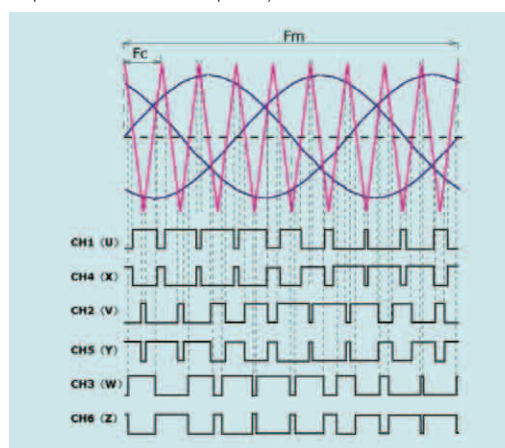
The pulse width, delay time, and other settings can be changed simultaneously for any combination of CH. Output example when the pulse width of channels 1 to 3 is changed simultaneously.

Signal generation method and output examples of the inverter option



Single-phase bipolar output in the INVERTER mode

Pulses can be easily generated by specifying the carrier frequency (F_c), modulation signal frequency (F_m), and modulation depth (that is, the rate of the modulation signal amplitude to the carrier amplitude).



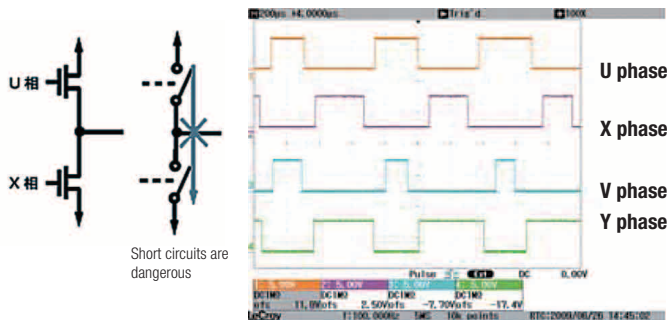
3-phase 2-level in the INVERTER mode

Pulses can be easily generated by specifying the carrier frequency (F_c), modulation signal frequency (F_m), and modulation period (that is, the rate of the modulation signal amplitude to the carrier amplitude).

Lineup

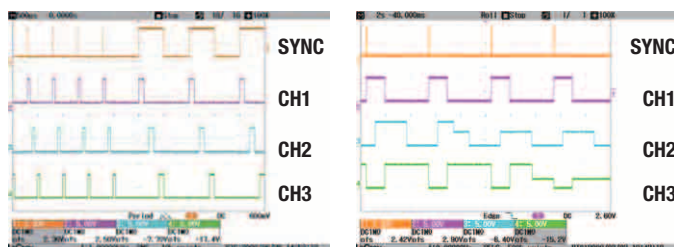
Items	Product name	Model number	Incorporated function
Main unit	Delay pattern generator	DG-8000	—
Software option	Inverter and PPG option	DG-801	INVERTER mode PPG mode
	Test adapter	DG-802	Operation pattern function
Hardware option	External modulation option	DG-601	External modulation function
	Quick synchronization option	DG-602	Quick synchronization function

Gap control to prevent the high and low side switches of devices from being turned on simultaneously



If the phase U and X devices in the above illustration are turned on at the same time, they short-circuit, causing danger and damage. The DG-8000 gap time control function automatically generates the specified dead time as shown in the illustration. Even if the frequency or cycle changes, the dead time remains constant. The gap time can be changed even during oscillation. It is also possible to turn devices on at the same time by specifying a negative value.

Independent control of the time axis and vertical axis



Changing the frequency

Changing the output level

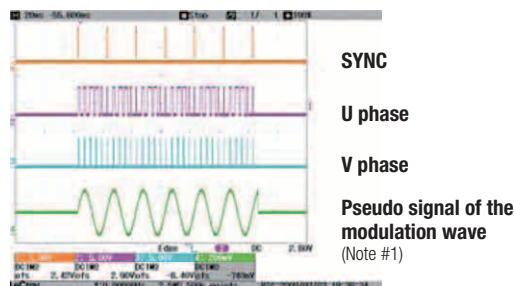
The parameters related to the time axis and those related to the vertical axis are separately controlled. These parameters can be changed manually or by using remote commands.

Support of ORed output on channel 1

Channel 1 has an ORed output function, which logically adds up to 6 sets of double pulses, making twelve pulses of specified channels, and outputs the result.

Easy generation of PWM signals

The inverter and PPG option (DG-801) enables you to output control signals for the buck chopper, single-phase uni-polar, single-phase bi-polar, and 3-phase 2-level. The modulation frequency and modulation depth can be changed even during oscillation. This is convenient for testing inverters because it is possible to obtain output to which pulse width modulation created from the inner sine wave and triangle wave is applied.

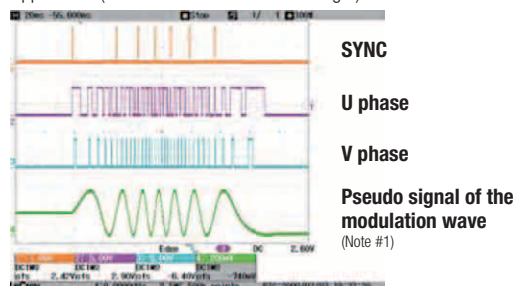


Note #1:
The modulation signal is shown only for explanatory purposes. This signal is not output from the DG-8000.
When using the PPG function, this generator functions as a signal generator for complicated logic modulation waves on 6 channels using predetermined pulse patterns. Waveform patterns can be created using the waveform creation application (which is available free of charge.)

Configuration example
DG-8000 main unit: 1
DG-801 inverter and PPG option: 1

Variable control of the PWM signal frequency

The operation pattern option (DG-802) is convenient for continuous operation testing because it enables variable control of the frequency and modulation depth (in the inverter mode only). The patterns for such control are controlled using predetermined arbitrary waveforms. These waveforms can be created using the waveform creation application (which is available free of charge.)



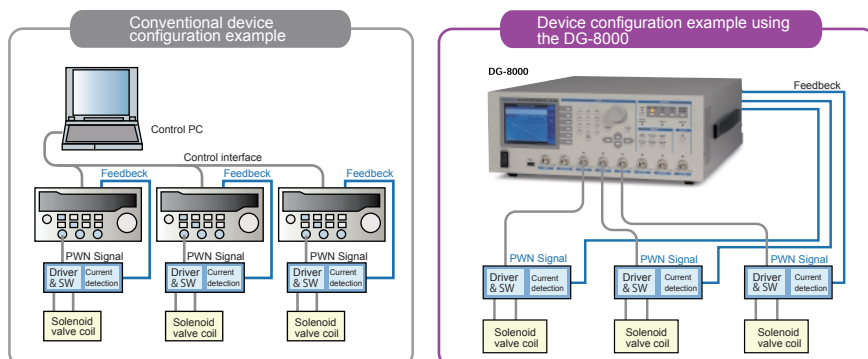
The illustration on the left shows an example of when a trapezoid waveform signal is used to apply frequency modulation.

Note #1:
The modulation signal is shown only for explanatory purposes. This signal is not output from the DG-8000.

Configuration example
DG-8000 main unit: 1
DG-801 inverter and PPG option: 1
DG-802 operation pattern option: 1

In the inverter mode, faulty patterns during the gap time can be inserted intentionally at regular intervals by using the error insertion function.

Application example: Continuous operation test of solenoid and other elements that control electromagnetic valves

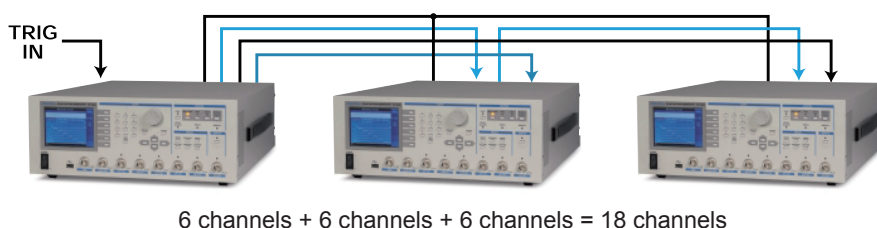


The external modulation option (DG-601) enables external control of the following functions:

- Modulation of the pulse width and delay in the basic mode
- Control of the modulation depth in the inverter mode
- Control of the frequency and modulation depth for operation patterns

Configuration example
DG-8000 main unit: 1
DG-802 operation pattern option: 1
DG-601 external modulation options

Parallel operation of three generators to support output from 18 channels



The quick synchronization option (DG-602) quickly enables up to 3 generators to synchronously operate by connecting BNC cables to the rear panel. If one of the generators goes down, the remaining two generators also shut down their output as a failsafe when this function is used.

Configuration example
DG-8000 main unit: 1
DG-602 quick synchronization options: 3

Delay Pattern Generator DG-8000 Specifications

Common specifications

Pulse output terminal

Number of channels	6CH
Output level	± 10V (open) / ± 5V (50 Ω)
Output range	2 ranges (large/small)
Output logic	Positive/negative
Output impedance	50 Ω
ORed output	Effective channels among channels 1 to 6 are ORed and the result is output (from channel 1)

Other output terminals

SYNC OUT output	BNC terminal (1)
IRREGULAR output	BNC terminal (1)
ALARM output	BNC terminal (1)
10 MHz REF output	BNC terminal (1)
REAR TRIG output	Quick synchronization operation option (DG-602), BNC terminal (1)

Input terminals

TRIG	BNC terminal (1), input: ±5V, max., threshold: ±1/2 of input level, variable
TRIG INH/RDY	BNC terminal (1), TTL level
Emergency stop input	BNC terminal (1), TTL level
10 MHz REF input	BNC terminal (1), 1V P-P ±100ppm or less required
Frequency control input	For the external modulation option (DG-601) and operation pattern option (DG-802), BNC terminal (1)
External modulation (PWM)	For the external modulation option (DG-601), BNC terminal (3)
REAR TRIG input	For the quick synchronization operation option (DG-602), BNC terminal (1)
ALARM SENSE input	For the quick synchronization operation option (DG-602), BNC terminal (1)

Output control

Oscillation start/stop	The button to turn all channels on or off immediately
Individual setting	To turn all channels on or off immediately
When oscillation stops	Select relay OFF or set the output level to 0.

LED indicators

TRIG'd	Indicates when TRIG is applied.
OUTPUT, channels 1 to 6	Indicates when output is enabled and on.
REMOTE	Indicates up in the REMOTE status.
INHIBIT/READY	Indicates up when oscillation is READY.

Pulse generation

Oscillation mode	CONT, TRIG'd CONT, TRIG, GATE
Gap control	Supported. *Gap control is a function that ensures non-overlapping time when phases V and X, phases U and Y, and phases W and Z overlap each other by specifying a delay or pulse width. This function can be also used to intentionally make these phases overlapped.

Interface

TRIG'd	USB1.1 storage function only (Waveform file and Setup file)
Remote (LAN)	100BASE-TX, 10BASE-T
Remote (GPIB)	Supported as standard

Screen display

LCD	4.7-inch color LCD
Resolution	320 x 240 pixels

Others

SETUP save/recall	Supported (10 internal memories)
Power-saving mode	Supported
Beep function	Supported
Status display	Supported

Power supply unit

AC power supply	AC 100V to AC 240V (50/60 Hz)
Power consumption	190VA, max

Mechanical section

External dimensions (mm)	Approx. 400 (W) x 150 (H) x 497 (D) (without external projections)
Weight	Approx. 8kg

Environment

Operating temperature	0°C to +40°C (without condensation)
Operating humidity	85% R.H. or less at +40°C
Storage temperature	-20°C to +60°C

Accessories

Power cable	1
Operation manual	CD-ROM (1)

The following modulations can be applied by using the DG-601 external modulation option when the main unit function is in the Basic mode:

PWM modulation

The pulse width can be changed by an external input signal. The modulation depth can be individually specified for each external input channel (U/V/W) and freely allocated to output channels. Delay modulation

The delay value can be changed by an external input signal. The modulation depth can be individually specified for each external input channel (U/V/W) and freely allocated to output channels.

Other specifications

BASIC mode	
Mode	Independent control of 6CH, 3-phase pattern A/B

6 independent channels

Number of pulses	SINGLE pulse/ DOUBLE pulse
Frequency/cycle	1mHz to 10MHz (1mHz or 9-digit resolution) 100ns to 1,000s (10ns or 9-digit resolution)
Frequency/cycle accuracy	±50ppm
Standard channel	Select SYNC or both edges of the smallest channel
Delay	0ns, 10ns to 1,000s (10ns or 9-digit resolution)
Pulse width	0ns, 50ns to 1,000s (10ns or 9-digit resolution)
PHASE	0° to 360° (minimum resolution: 0.01°, frequency-dependent) 0% to 100% (minimum resolution: 0.001%, frequency-dependent)
DUTY	0° to 360° (minimum resolution: 0.01°, frequency-dependent) 0% to 100% (minimum resolution: 0.001%, frequency-dependent)
Gap time setting	0 to ±1 cycle or 1s, max.
Gap resolution	Frequency specifying : Gap in 20 ns or 6 digits Cycle specifying : Gap in 10 ns or 6 digits
Frequency dividing function	Supported
Frequency dividing setting range	1 to 65,535
Tracking	Multiple parameters can be changed simultaneously.
Internal modulation	PWM modulation and delay modulation

3-phase pattern A

Oscillation mode	CONT, TRIG'd CONT, GATE
Cycle (Tc)	Determined by setting Tw1 and Tw2. Tc = (Tw1+Tw2) x 3
Tw1 and Tw2 setting range	0ns, 100ns to 100s
Tw3 setting range	0ns, 100ns or more (Fc minus- Tw1)
Pulse width setting resolution	100ns or 9digits
Gap control	By setting Tw3.
Operation change during oscillation	Parameters can be seamlessly changed.

3-phase pattern B

Oscillation mode	CONT, TRIG'd CONT, GATE
Cycle (Tc)	Determined by setting Tw and Tw3. Tc = Tw2+Tw3
Tw1 setting range	0ns, 100ns to 100s
Tw2 setting range	0ns, 100ns or up to more (Fc-2 x Tw1)
Tw3 setting range	100ns to 100s
Pulse width setting resolution	100ns or 9digits
Gap control	Realized by setting Tw2.
Operation change during oscillation	Parameters can be seamlessly changed.

Inverter mode (with the DG-801 inverter and PPG option mounted)

Mode	Buck chopper, single-phase uni-polar, single-phase bi-polar 3-phase 2-level
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Common setting parameters

Carrier frequency	100mHz to 1MHz
Modulation frequency	1mHz to 10kHz
Other parameters	Modulation depth, modulation steps, gap time, and others

PPG mode (with the DG-801 inverter and PPG option mounted)

Frequency specifying mode	
Frequency	1mHz to 10MHz (1mHz or 6-digit resolution)
Memory length	10kW or 100kW

Clock specifying mode

CK frequency	100Hz to 100MHz (resolution: 1mHz or 6digits)
Memory length	10kW or 100kW

Operation pattern (with the DG-802 operation pattern option mounted)

Frequency control	The frequency (cycle) can be controlled using any waveform or external input.
Frequency control input	BNC terminal (1)
Modulation control	INVERTER mode only. The modulation can be controlled using any waveform or external input.
Faulty pattern insertion	Supported

External modulation (with the DG-601 external modulation option mounted)

External modulation input	BNC terminal (3)
Frequency control input	BNC terminal (1)
Input range	2 ranges (-2 to +2V or 0 to +2V)
Input impedance	Approx. 1MΩ
Resolution	12 bits
Frequency characteristics	100kHz, amplitude of 90% or more (1kHz standard)

External modulation (with the DG-601 external modulation option mounted)

REAR TRIG output	BNC terminal (2)
REAR TRIG input	BNC terminal (1)
ALARM SENSE input	BNC terminal (1)

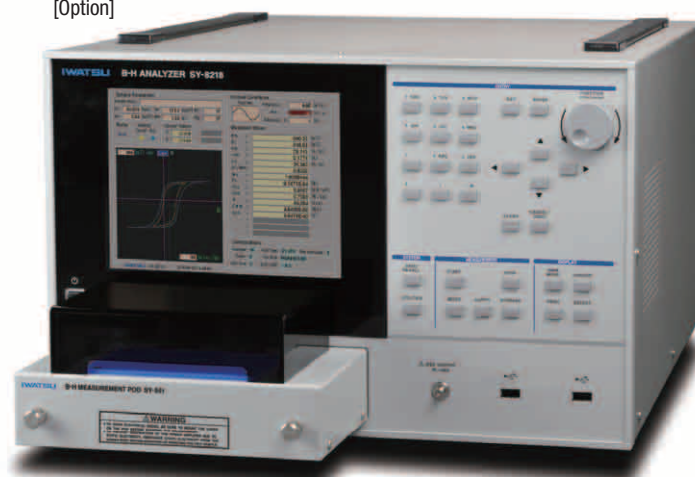
B-H Analyzer SY-8210 Series

SY-8218 10Hz - 10MHz

SY-8219 10Hz - 1MHz

Best suited for measuring magnetic properties of soft magnetic materials such as Silicon-steel plates, ferrites, and amorphous materials


 [Option]

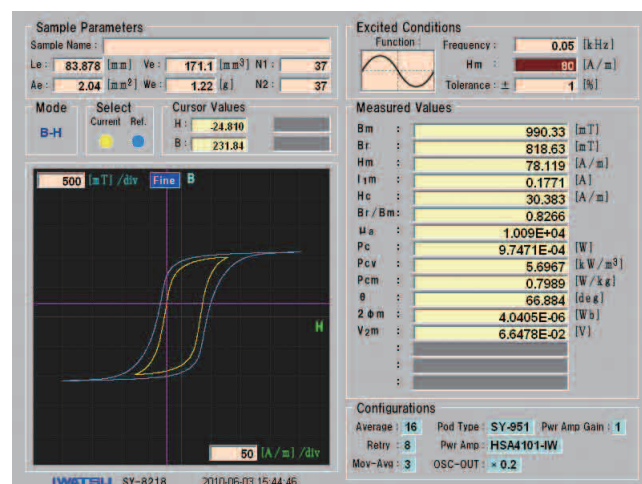


B-H Analyzer SY-8218


 [Option]



B-H Analyzer SY-8219



Main features

- Wide measurement frequency for materials analysis which used in high frequencies
 SY-8218: SINE from 10Hz to 10MHz, Pulse at Duty 50:50 from 10Hz to 1MHz
 SY-8219: SINE from 10Hz to 1MHz, Pulse at Duty 50:50 from 10Hz to 1MHz
- 16times of acquisition data(comparing with current Iwatsu models)
 Acquisition data at 8,192points/cycle perform precise measurement on parameters such as Hc(Coersive force), Br(Residual flux density), and other parameters.
- Pulse excitation function
 Both SINE(Sinusoidal) and Pulse(at Duty 50:50, 1MHz,max.) excitation are available as a standard function
- Temperature characteristic test with Scanner Chamber System
- Optional Items
 - Power Amplifiers
 - DC bias power supply*
 - Single plate measurement function*
 - High-current POD*

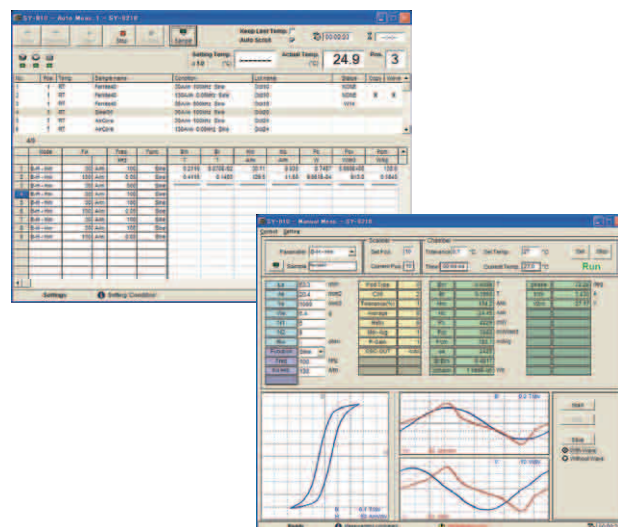
*under development

Model	SY-8218	SY-8219
Measurement method	CROSS-POWER method (conformance to IEC62044-3)	
Measurement mode	B-H measurement, Pc measurement, μ measurement	
Measurement item	Characteristics value	
	B-H measurement: Max. magnetic flux density (Bm), residual magnetic flux density (Br), max. magnetic field (Hm), coercive force (Hc), squareness ratio (Br/Bm), amplitude permeability (μ_a), core loss (Pc, Pcv, Pcm), current/voltage phase (Θ), total magnetic flux change ($2\phi_m$)	
	Pc measurement: Max. magnetic flux density (Bm), residual magnetic flux density (Br), max. magnetic field (Hm), coercive force (Hc), amplitude permeability (μ_a), core loss (Pc, Pcv, Pcm), phase (Θ), apparent power (VA)	
	μ measurement: Max. magnetic flux density (Bm), max. magnetic field (Hm), impedance permeability (μ_z), complex permeability (μ , μ''), amplitude permeability (μ_a), core loss (Pc), phase (Θ), loss coefficient ($\tan\delta$), inductance (L), resistance (R), impedance (Z), quality factor (Q)	
Waveform display	B-H curve, waveforms of excitation current, induced voltage, magnetic field, magnetic flux density	
Measurement frequency	10Hz to 10MHz	10Hz to 1MHz
Magnetic field signal detection	Voltage drop at both edges of non-inductive resistor Maximum signal detection current: 6A	
Magnetic flux density signal detection	Voltage detection at both edges of induced voltage detection coil Maximum signal detection voltage: ± 200 V	
Digitizer resolution	16 bits (8,192 points/cycle)	
Sample connection method	2 or 1 coil (winding) method	
Display	Color LCD display (800 x 600 pixels)	
Power supply	AC100V to 240 V, 50/60Hz, Power consumption: Approx. 130VA MAX	
Weight and dimensions	Main unit: Approx. 12.5kg, 420 (W) x 266 (H) x 480 (L) $\pm$ 2mm (without the projection section)	
Interface	USB (data storage)	

SY-810

- Enables auto-program measurements for temperature characteristics in combination with constant-temperature scanner systems.
- Auto-program measurements also possible with just the B-H analyzer.

- A maximum of twenty temperature conditions and a maximum of forty excitation conditions can be set for a single test sample, which enables 800 (= 20 x 40) different types of programming for the measurement conditions.
- Pulse excitation for the B-H analyzer can also be remotely controlled.
- Hard copies of the B-H analyzer measurement screen can also be automatically saved onto a USB memory in the JPEG or PNG formats.



The diagram illustrates the recommended USB-GPIB interface setup for the SY-320/321 chamber. It shows the following components and connections:

- PC:** A laptop running SY-810 remote control software, connected to the B-H Analyzer via a USB cable.
- B-H Analyzer SY-8218/8219:** The central control unit, connected to the PC via USB and to the Power Amp via an OSC cable (BNC-SMA).
- Power Amp:** A power amplifier unit with an INPUT (BNC) and an OUTPUT (BNC). It is connected to the B-H Analyzer's OSC OUTPUT (SMA) and to the Chamber SY-320/321 via a Power amp cable (BNC-BNC).
- Chamber SY-320/321:** The main system unit, which receives B&H signals from the Power Amp's output.
- USB-GPIB Interface:** A SY-509 USB-GPIB converter is connected to the Chamber SY-320/321 via a GPIB cable (408JE-101 GPIB cable) and to the B-H Analyzer via a USB cable.

The overall connection path for the B-H signals is: PC → B-H Analyzer → Power Amp → Chamber SY-320/321.

USB - GPIB interface: NI GPIB-USB-HS hi-speed GPIB controller by National Instruments recommended.

[182766-01 bulkhead adapter manufactured by National Instruments](#)

The diagram illustrates the connection between a PC and a power amplifier via a USB-GPIB interface. A laptop on the left is connected to a B-H Analyzer (SY-8218/8219) via a USB cable. The B-H Analyzer is connected to a SY-509 USB-GPIB converter, also via a USB cable. The SY-509 converter is connected to a Power Amplifier. The Power Amplifier has four ports: POWER INPUT (BNC), OSC OUTPUT (SMA), INPUT (BNC), and OUTPUT (BNC). The B-H Analyzer is connected to the POWER INPUT (BNC) of the Power Amplifier via a POWER cable (BNC-BNC). The SY-509 converter is connected to the INPUT (BNC) of the Power Amplifier via a BNC cable. The Power Amplifier has an OSC OUTPUT (SMA) port connected to the B-H Analyzer via an OSC cable (BNC-SMA). The Power Amplifier has an OUTPUT (BNC) port connected to the B-H Analyzer via a Power amp cable (BNC-BNC).

PC
SY-810 remote control software

USB

B-H Analyzer
SY-8218/8219

SY-509 USB-GPIB converter

USB

Power Amplifier

POWER INPUT (BNC)

OSC OUTPUT (SMA)

INPUT (BNC)

OUTPUT (BNC)

POWER cable (BNC-BNC)

OSC cable (BNC-SMA)

Power amp cable (BNC-BNC)

Main Options and System Examples

Constant-temperature Scanner System Options

Constant-temperature Chamber scanner system

SY-320
SY-321

For evaluations of samples' magnetic characteristics vs temperature

Automatic measurements possible with the SY-810 (software.)

Temperature Range		-30°C to +150°C
Sample Quantity	SY-320	20pcs
	SY-321	41pcs
Maximum Measurement Current		6Apk



SY-321

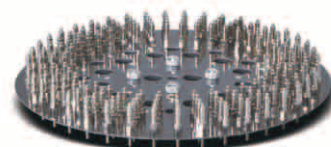
Spare turntable

SY-510
SY-511

A table for mounting samples

	Scanner	Number of samples
SY-510	SY-320	20pcs
SY-511	SY-321	41pcs

*Supplied with the SY-320 x 1 unit and SY-321 x 1 unit as standard



SY-511

Chamber for measuring constant-temperature magnetic characteristics

IE-1210

High-Temperature Constant-temperature Chamber

(Ambient temperature +20°C to +300°C)

Spare contact pin set

SY-512

Consumable Components for Maintenance Purposes

Consumable products for carrying out maintenance on constant-temperature scanner systems SY-320 and SY-321.

*Supplied 1set (4pcs) with the SY-320 and SY-321 as standard



Software Options

PC Software

SY-810

Remote Control Software

(Supplied with the SY-509 and bulk head adaptor as standard accessories.)

The NI GPIB-USD-HS manufactured by National Instruments is required separately for PC connection purposes. (see page34)



Interface

NI GPIB-USB-HS
GPIB Controller Supporting
Hi-Speed USB

A USB-GPIB converter for use with controllers connected to PCs when the SY-810 remote control software is being used.

Interface

SY-509
USB-GPIB Conversion Adaptor

*Supplied with the SY-810 as standard equipment.



Power Amplifiers

Model		Frequency	Output Current	Output Voltage	Power Consumption
HSA4101-IW		DC to 10MHz	± 1A(MAX)	± 71V(MAX)	50VA(MAX)
HSA4014-IW		DC to 1MHz	± 5.6A(MAX)	± 75V(MAX)	200VA(MAX)
IE-1125		DC to 3MHz	± 5.2A(MAX)	± 140V(MAX)	350VA(MAX)

Example of Full-automatic B-H Analyzer with Constant Temperature Scanner System for various evaluations

Temperature range : -30°C to +150°C

Max. number of samples 20 : SY-320

Max. number of samples 41 : SY-321

Remote control PC software SY-810(option)

Control PC



Power Amplifier IE-1125

350VA, 140V, 5A



B-H Analyzer SY-8219

10Hz-1MHz



Constant Temperature Chamber Scanner System

SY-321 for Max. 41 samples

Temperature range :
-30°C to +150°C
(SY-320: 20samples)

Empty toroidal coil

SY-513

A toroidal-shaped empty case. It is used for measurement of a powder sample.



IWATSU

<http://www.iti.iwatsu.co.jp/>

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